

FORECASTING CONDITIONAL CLIMATE-CHANGE USING A HYBRID
APPROACH

By

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B.S., University of Colorado Denver, 2010

A thesis submitted to the
Faculty of the Graduate School of the
University of Colorado in partial fulfillment
of the requirements for the degree of
Masters of Integrated Sciences
Integrated Sciences

2013

This thesis for the Masters of Integrated Science degree by

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September 23, 2013

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Modeling Climate-Change in the Southwestern United States

Thesis directed by Professor Weldon Lodwick.

ABSTRACT

A novel approach is proposed to forecast the likelihood of climate-change across spatial landscape gradients. This hybrid approach involves reconstructing past precipitation and temperature using the self-organizing map technique; determining quantile trends in the climate-change variables by quantile regression modeling; and computing conditional forecasts of climate-change variables based on self-similarity in quantile trends using the fractionally differenced auto-regressive integrated moving average technique. The proposed modeling approach is applied to states (Arizona, California, Colorado, Nevada, New Mexico, and Utah) in the southwestern US, where conditional forecasts of climate-change variables are evaluated against recent (2012) observations, evaluated at a future time period (2030), and evaluated as future trends (2009-2059). These results have broad economic, political, and social implications because they quantify uncertainty in climate-change forecasts affecting various sectors of society. Another benefit of the proposed hybrid approach is that it can be extended to any spatiotemporal scale providing self-similarity exists.

The form and content of this abstract are approved. I recommend its publication.

Approved: Weldon Lodwick

DEDICATION

I dedicate this work to the tireless support of my loving wife Naimeh and to my parents who made the biggest sacrifice by leaving behind their family so that we could have a better life.

ACKNOWLEDGMENTS

I would like to thank Michael J. Friedel, my teacher, mentor, and friend. For without his tireless push, this research would have not been possible

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1. Introduction

People benefit from a multitude of resources and processes supplied by natural ecosystems (Randhir and Ekness, 2009). These benefits include water resources suitable for supporting various sectors of society such as agriculture, construction, daily living, energy, fishing, forestry, manufacturing, public health, recreation, transportation, and overall economic development that maintains life systems prompting sustainability. Climate change is frequently cited as one external driver of ecosystems (Furnis, 2010). Because climate is temporally and spatially dependent, change at a global scale differs from regional or local scales (Friedel, 2012b). One reason for spatial differences is the superposition of large-scale climate patterns due to atmospheric and oceanic teleconnections (Schwing et al., 2002). Climate change also differs across temporal scales over which there are variations in amplitude, gradient, and duration (Woodhouse and Overpeck, 1998; Mann et al., 2008; Mann et al., 2009; Friedel, 2011a, Friedel, 2012a).

In many studies, the duration of climate change is considered short-term (years to decades) variability (Woodhouse and Overpeck, 1998, Ubilava and Helmers, 2013). Short-term climate variability is often attributed to oscillations in the sea surface temperature (SST) that alter ocean currents and overlying air pressure resulting in a redistribution of temperature and precipitation (Smith and Reynolds, 2003). The El Niño Southern Oscillation (ENSO) is considered the strongest short-term periodic fluctuation (2-7 years) with a rise (El Niño) or decrease (La Niña) of SST in the equatorial Pacific Ocean (Blade et al., 2008, Ubilava and Helmers, 2013). The influence of this teleconnection is not uniform in the United States, and ENSO events can affect things like water supply, water quality, riparian habitat, power generation, and range productivity. Related drought

consequences often include crop failure, debris flows, insect infestations, pestilence, violent conflict, wildfires, and disruptions to economic and social activities (Riebsame et al., 1991). Long-term climate variability (hundreds to thousands of years) is often attributed to alterations in geologic and extraterrestrial processes, such as volcanic aerosols (Rampino and Self, 1982) and solar activity (Gray et al., 2010). Long-term climate-change reconstructions provide insight on past surface temperature and drought variability at timescales crossing centuries or millennia (Mann et al., 2009; Friedel, 2012a). Should natural or anthropogenic forcing influence the frequency or intensity of climate change, there is an increased likelihood for future ENSO hazards placing national and global security at risk (Riebsame et al., 1991).

For these reasons, climate-change forecasts could benefit many sectors of society, but the scale-dependent complexities render it a challenge using traditional process-based models. Specifically, climate forcing is known to interact with ecosystems characterized by coupled, nonlinear, and multivariate processes. Data associated with these ecosystems are typically sparsely populated ranging spatially from local (1000s km²) to global and temporally from immediate (1-10s years) to long-term (100s to 1000s years). One critical issue is the lack of essential calibration data that results in large inaccuracies (Loke et al., 1999). Other nonprocess-based modeling efforts include linear time-series models, such as the Autoregressive (AR) or Autoregressive Integrated Moving Average (ARIMA) models (Said and Dickey, 1984; Cowpertwait and Metcalfe, 2009). These traditional linear time-series modeling schemes are too rigid with respect to detecting unexpected features like the onset of nonlinear trends, or patterns restricted to sub-samples of a data set.

One alternative paradigm is to use a hybrid approach in which soft computing provides data for subsequent use in traditional numerical or empirical modeling. Some applications of the hybrid modeling approach are in rainfall-runoff (Jain and Kumar, 2007), debris-flow (Friedel, 2011b), mineral-resource (Friedel, 2012b), and unexploded ordinance (Friedel et. al., 2012). In this study, the goal was to evaluate the efficacy of hybrid modeling for forecasting climate-change over spatial landscape gradients in the southwestern United States. The objectives were to: (1) evaluate independent observations against the 2012 forecast of temperature, precipitation, and drought across California, Colorado, and Nevada; (2) evaluate future forecast trends over 50 years of probable temperature, precipitation, and drought across Arizona, California, Colorado, New Mexico, Nevada, and Utah; and (3) to evaluate the performance characteristics of models that are generated (Bennett et al, 2013). However, in today's modeling world, it is not just enough to evaluate the performance of a model, but it is a necessity to account for model uncertainty. It is to this end that this study introduces a new modeling paradigm to account for uncertainty in time-series forecasting.

This study extends the work of Friedel (2011a) who sought to reconstruct 2,000 years of past temperature and precipitation for the south-central and southwestern United States; Esfahani and Friedel (2010) who identified a long-memory process in reconstructed climate variables; Friedel (2012a) who used quantile regression to quantify uncertainty in global reconstructions of past temperature and precipitation and Caballero et al. (2002) and Nunes et al. (2011) who used a fractal approach to investigate the long-memory process associated with temperature.

2. Methodology

In modeling climate-change variables, the hybrid approach relies on four computational steps: reconstruction, trends, forecasts, and uncertainty (Fig. 2.1). Each of these steps is briefly described in the following sections.

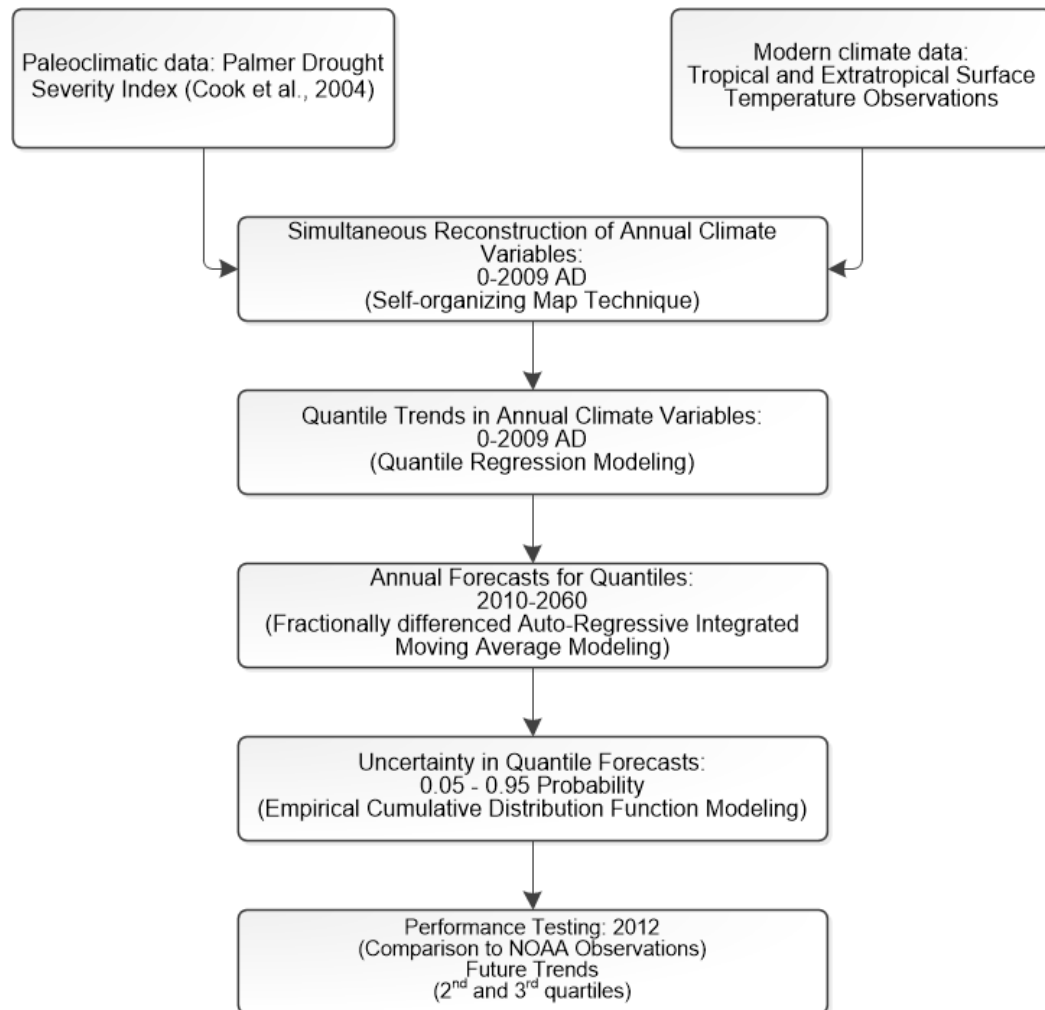


Fig. 2.1- Schematic depicting the hybrid-modeling framework used to forecast climate-change.

2.1 Reconstructing climate-change variables

The reconstruction of climate-change variables (temperature and precipitation) follows the approach described by Friedel (2012a). In that approach, a self-organizing map (SOM) technique was used to project input data to a discrete lattice of competitive

neurons (Kohonen, 2001), and estimates past climate-change variables by minimizing topological error vectors (Fraser and Dickson, 2007). The process of projecting data is essentially a data compression technique (Hastie et al., 2002) for which the success of topology-preservation was analyzed based on the quantization error $E(G, X)$, given by:

$$E(G, X) = \frac{1}{N} \sum_{i \in Q} \sum_{j=1}^M h_{i,j} \|x_j - w_i\|^2, \quad (1)$$

where w_i are weight vectors assigned to a fixed number of N neurons in the map grid G , x_j are the M input data vectors (economic mineral-resource variables), $h_{i,I}$ is a neighborhood function, $\|x_j - w_i\|$ is the Euclidian norm, and I is the best matching unit (BMU) vector.

Implementation of the SOM learning method is based on the stochastic gradient described by Kohonen (2001). It consists of a two-step process that is performed each time an input pattern is presented to the map: competition to determine the BMU and cooperative learning (spreading information contained in the current input vector across the map). At the beginning of the unsupervised training phase, the weight vectors are initialized to small random numbers. The input data vectors are presented to the map grid in a random fashion to generate data clusters without introducing bias for a specific class. In the first step, the BMU with map coordinates (I_i, I_j) is determined as the grid neuron, whose weight vector is the closest to the input given by:

$$I = \text{ArgMin}_{i,j \in G} \|x(i) - w(j)\|, \quad (2)$$

where ArgMin is the minimum distance defining the central position of the neighborhood function. The neighborhood function $h_{i,I}$ is chosen to be Gaussian function given by:

$$h_{i,I}(n) = \exp \left[-\frac{\|r_i - r_I\|^2}{\sigma(n)^2} \right], \quad (3)$$

where $\|r_i - r_I\|$ corresponds to the distance between map neuron r_i and BMU in the map grid, and $\Phi(n)$ defines the width of the neighborhood function, a monotonically decreasing function of the iteration (also called epoch) number n . In the second step, a weight update is determined which is a function of the distance to the current BMU, as expressed through the neighborhood function $h_{i,I}(n)$. The weights are gradually adjusted according to:

$$w_i(n+1) = w_i(n) + \alpha(n)h_{i,I}(n)[x(n) - w_i(n)], \quad (4)$$

where $\alpha(n)$ is a scalar value called the learning rate bounded on the interval $[0, 1]$. The BMU ensures that the largest weight correction ($h_{i,I}(n)=1$) is adjusted in the direction of the input vector. The association effect takes place at the neighboring nodes but to a lesser degree because of the Gaussian shape. This adaption procedure stretches the weight vectors of the BMU and its topological neighbors towards the input vector. Presenting similar input vectors to the map provides further activations in the same neighborhood and thereby tends to produce clustering of data in the feature space. Association between neurons decreases during the learning process (the width of the neighborhood function $\Phi(n)$ is forced to decrease with n preserving large clusters of data while enabling the separation of clusters that are closely spaced). Ultimately, this training process results in a topology where similarities among data patterns are mapped into similar weights of the neighboring neurons, and the asymptotic local density of the weights approach that of the training set (Ritter and Schulten, 1986).

Cross-validation (Efron and Tibshirani, 1993) is conducted to ensure the SOM provides unbiased estimates of climate-change variables. In this case, known data values are estimated based on distances among the available model vectors (Wang, 2003; Kalteh and Berndtsson 2008). In the traditional approach, estimates of values are taken directly from the prototype vectors of the best matching units (Fessant and Midenet, 2002; Wang, 2003). Often times certain training data sets result in biased estimates (Dickson and Giblin, 2007; Malek, et al., 2008) requiring a modified scheme that incorporates bootstrapping (Breiman, 1996), ensemble average (Rallo et al., 2002), or nearest neighbor (Malek et al., 2008). This study uses an alternative iterative estimation scheme that minimizes the topological error vector (Fessant and Midenet, 2002). The estimation of past climate-change values for all variables is done simultaneously and referred to here as the reconstruction. For more details about SOM training and estimation, the reader also is referred to (Kohonen, 2001; Vesanto, and Alhoniemi, 2000).

2.2 Quantile trends in climate-change variables

The determination of quantile trends in the reconstructed climate-change variables follows the approach described by Friedel (2012). One advantage of this approach is its flexibility in modeling data with conditional functions that may have systematic differences in dispersion, tail behavior, and other covariate features (Koenker, 2005). In adopting this approach, the quantile regression approach is adopted thereby reducing the d-dimensional nonparameteric regression problems to a series of additive univariate problems given by:

$$\min_{\alpha \in \mathbb{R}} \left(\sum_{i=1}^n \rho_i(y_i - \alpha_i) - \lambda \sum_{j=1}^{n-2} |d_j^T \alpha| \right), \quad (5)$$

where $\min$ is the minimization operator and α_i estimated coefficients at distinct x_i . d_j is an n -dimensional vector with elements $(h_j^{-1}, (h_{j+1}^{-1} + h_j^{-1}), h_{j+1}^{-1})$ in the j , $j+1$, and $j+2$ positions and zeros elsewhere. $h_i = x_{i+1} - x_i$; ρ_i is the quantile regression loss function; λ is a factor that controls the degree of smoothing; T is the transpose; x_i are response observation values associated with random variable X ; and y_i are response observation values associated with random variable Y .

Individual quantile curves can then be specified as a linear b-spline of the form:

$$Q_{Y_i}(\tau|Y_{t-1}) = \sum_{i=1}^p \phi_i(Y_{t-1}) \beta_i(\tau), \quad (6)$$

where $Q_{Y_i}(\tau|Y_{t-1})$ is the τ th conditional quantile function, $\beta_i(\tau)$ are the regression quantiles; $\beta_i = \alpha_{i+1} - \alpha_i/h_i$, are regression coefficients; $\phi_i(\cdot): i = 1, \dots, p$ denote the basis function of the spline; and τ is the quantile ($0 < \tau < 1$). Once the knot positions of the spline have been selected, such models are then linear in parameters and therefore can be estimated (Koenker, 2005).

2.3 Forecasts of climate-change variables

2.3.1 Test of determinism

Before fractal techniques can be used to analyze the reconstructed climate-change data, the determinism of a fractal time series must be established. The data must be identified as deterministic or statistical (Kaplan, 1994; Turcotte, 1997). The other assumption to be satisfied is that stationarity exists for long-memory process data (Beran,

1994). The determinism test developed by Kaplan (1994), allows the determination as to whether a time series has a deterministic or stochastic structure. Turcotte (1997) defines the set as a deterministic fractal set, if the set is scale invariant at all scales; and a statistical fractal set, if the set is different at different scales but the differences do not allow the scale to be determined. This is an important distinction because it determines the correct way to calculate the fractal dimensions (as defined by equation (10)) for a time series.

Mathematically, the delta-epsilon test of continuity (Kaplan, 1994) is applied to determine the deterministic structure of a time series. To do so, the continuity test is applied to orbits comprising the phase space topology created by time-delayed embedding of the original set. This process is facilitated by generating an ensemble set of surrogate time series using a bootstrapping approach (Constantine et al., 2010). Next, the phase-space statistic, called the E-statistic (Kaplan, 1994), is calculated for the time-delayed embedding of the original time series as well as the ensemble of surrogate data. The structure is then judged by comparing the E-statistics of the original set to the E-statistic of the surrogate set. A separation between the two statistics implies the existence of a deterministic structure, and the converse implies the set is a realization of a random process and thus has a stochastic structure.

According to Kaplan (1994), the E-Statistic is defined as:

$$\delta_{j,k} = |z(j) - z(k)|, \quad (7)$$

$$\epsilon_{j,k} = |z(j + \kappa) - z(k + \kappa)|, \quad (8)$$

$$\varepsilon(r) = \overline{\epsilon_{j,k}} \text{ for } j, k \text{ s. t. } r \leq \delta_{j,k} < r + \Delta r, \quad (9)$$

where $\delta_{j,k}$ is the Euclidean distance between phase space points $z(j)$ and $z(k)$, and $\epsilon_{j,k}$ is the corresponding separation distance between the points at a time κ points in the future along their respective orbits. The variable κ is the orbital lag and the future points are the images of the original pair. The increment Δr is the width of a specified Euclidean bin size. Given Δr , the distance $\delta_{j,k}$ is used to identify the proper bin in which to store the image distance $\epsilon_{j,k}$ and the average of each bin forms the $\epsilon(r)$ statistic. Finally, the E-statistic is formed by calculating the cumulative sum of the $\epsilon(r)$ statistic (Constantine and Percival, 2011).

2.3.2 Determining the fractal dimension

Fractal sets are characterized by their dimensions (Mandelbrot, 1967; Mandelbrot and van Ness, 1968). Fractal dimensions are mathematically different from topological dimensions in that while topological dimensions are strictly constructed from integers, fractal dimensions can be fractional. While there are an infinite number of fractal dimensions, the fractal information dimension is important because it describes the entropy of a data set. In general, the fractal dimension is described by:

$$D_q = \frac{1}{q-1} \frac{\partial \log \sum_i p_i^q}{\partial \log(r)}, \quad (10)$$

where q refers to the dimension of interest. By embedding the dataset in an n -dimensional grid with cells having sides of size r , we compute the frequency for which a data point falls into the i^{th} cell, p of I . The information dimension is described by $\lim_{q \rightarrow 1} D_q$. As the information dimension approaches 1, the numerator of equation (4) changes to the Shannon's entropy describing changes in the entropy and trend of a time series (Barbara,

1999). Under these conditions, the fractal dimension of a stochastic self-similar data set simplifies to:

$$D_q = \frac{1}{q-1} \frac{\log \sum_i p_i^q}{\log(r)}, \quad (11)$$

Without the partial differential equation requirement the problem becomes computationally tractable and can be calculated easily, while there are several ways to calculate the fractal dimensions, for the purposes of this paper, the box counting technique is used (De Pison et al., 2008).

2.3.3 Long-memory process

The existence of a long-memory process was first explored by Hurst (1951) when trying to find a solution on how to regularize flow of the Nile River. He observed that long periods of high-flow levels were followed by long periods of low-flow levels indicating the existence of a long-memory process. In general, long-memory process describes the process where by a time series shows that past values have a strong effect on present or future values. Mandelbrot and van Ness (1968) introduced the Hurst parameter (H) to describe the long-term memory of a time-series process. The Hurst parameter is given by:

$$H = 1 - \frac{\lambda}{2}, \quad (12)$$

where λ is the long-memory parameter with the range of $0 < \lambda < 1$; thus, H has a range of $\frac{1}{2} < H < 1$. The closer H is to 1, the more persistent the time series is considered and at values less than or equal to $\frac{1}{2}$, a long-memory process does not exist. Currently there are several methods to estimate the Hurst parameter; however, Rae et al. (2011) showed

that Whittle's maximum likelihood estimation approach produced the best results (Beran, 1994).

2.3.4 Fractionally differenced auto-regressive integrated moving average model

If climate-change variables have a long-memory process then the fractionally differenced auto-regressive integrated moving average model (FARIMA) can be applied and forecasts generated. The FARIMA model is similar to the Box-Jenkins ARIMA model (Cowpertwait and Metcalfe, 2009) except that the integrated part of the ARIMA model can be a fractional number defined (Beran, 1994) as:

$$d = H - \frac{1}{2}, \quad (13)$$

This expression underscores the need for a good estimation of the Hurst parameter when dealing with climate time-series data. If the climate-change data exhibits long-memory process and stationarity, the FARIMA model assumptions are satisfied and can be used to generate forecasts following three steps. First, the model parameters for each time series are determined using the maximum-likelihood estimators (Fraley et al., 2011). Second, the model is fitted and forecasts made using the estimated parameters (Hyndman, 2011). Third, the forecasts are validated using various statistical measures (Hyndman and Koehler, 2006).

2.3.5 Empirical cumulative distribution function

The use of quantile regression facilitates quantization of the prediction uncertainty by constructing empirical cumulative distribution functions (ECDF) at any year of a forecast. The empirical cumulative distribution function (ECDF) is determined by modeling the collection of quantile forecast results. The ECDF is a step function F_n :

$$F_n(t) = \frac{\{x_i \leq t\}}{n} = \frac{1}{n} \sum_{i=1}^n 1(x_i \leq t), \quad (14)$$

where i is the number of tied observations at that value (missing values are ignored). For observations $x = (x_1, x_2, \dots, x_n)$, and F_n is the fraction of observations less or equal to t .

2.4 Software for reconstruction, quantile trends, and forecasting and empirical cumulative distribution functions

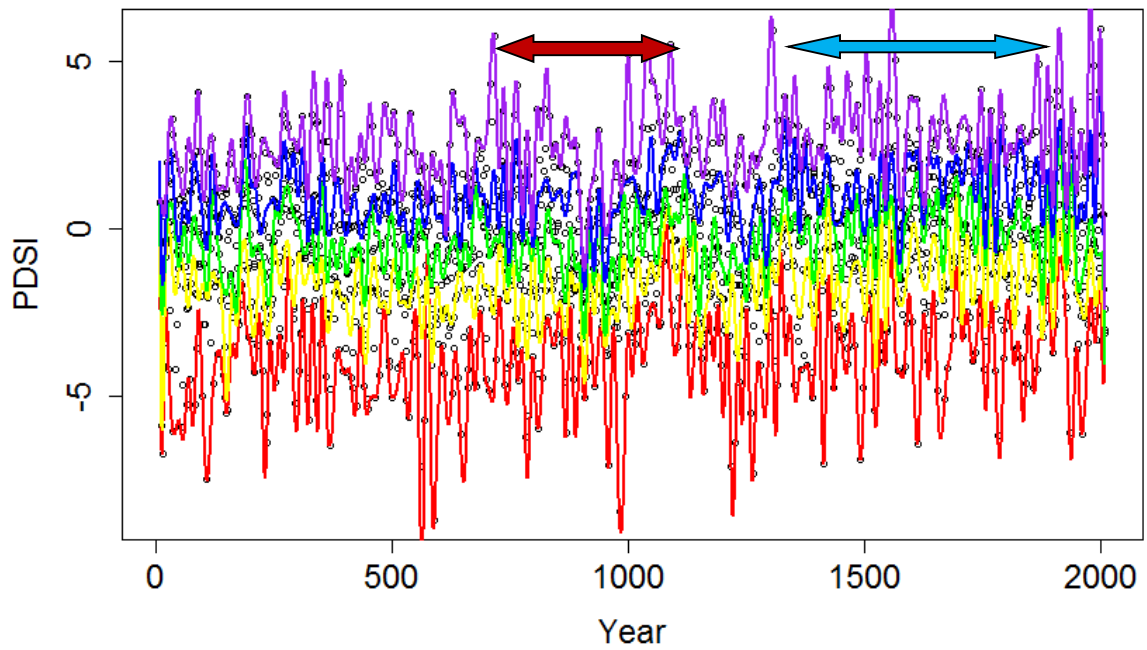
The data mining, reconstruction, and analysis are carried out using the SiroSOM (CSIRO Exploration & Mining, 2008) graphical user interface (GUI). This GUI provides an interface between data sets and functions in the freely available SOM Toolbox (Adaptive Informatics Research Center, 2010). Quantile modeling is conducted using the quantreg, splines, and stats packages (Hornik, 2011); fractal transformation and fractal dimension calculations are conducted using the fdim package (Martinez de Pison, Ascacibar et al. 2008). Fractionally differenced auto-regressive integrated moving average modeling is conducted using the forecast and fractal package (Constantine and Percival, 2011, Hyndman 2011) and the empirical cumulative distribution function modeling is conducted using the stats package (Hornik, 2011); freely available at <http://www.r-project.org/> in the R toolbox.

3. Results

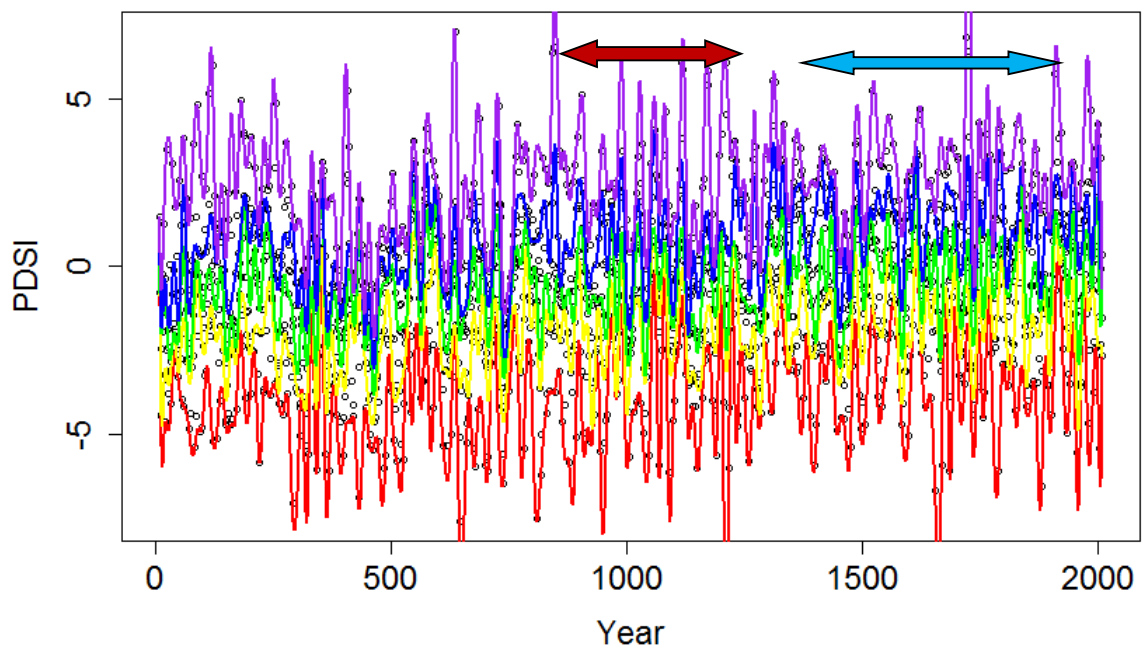
3.1 Reconstructing climate-change variables

Annual temperature and precipitation values were reconstructed across a gradient of modern climate zones: Arizona (Desert), California (Mediterranean), Colorado (Semiarid to Alpine), Nevada (Semiarid to Arid), New Mexico (Semiarid), and Utah (Semiarid). The simultaneous reconstruction of annual temperature and precipitation was done based on the self-organized nonlinear data-vector relations among approximately 2,000 years (0 to 2009 AD) of reconstructed warm-season (average of June, July, and August) Palmer Drought Severity Index (PDSI) data (Cook et al., 2004), 114 years (1895–2009) of annual state precipitation (accumulation over January through December) and temperature (average of January through December) data (National Climatic Data Center, 2010), and other related tropical and extratropical measurements described by Friedel (2012a). The temperature and precipitation are standard climate variables, whereas the PDSI defines annual dry, neutral, and wet periods based on tree-ring information (Palmer, 1965; Cook et al., 2004). For a comprehensive review of the reconstruction and validation of these data, the reader is referred to the cited references. In this study, the reconstructions were verified against independent precipitation and temperature data for the years: 1896, 1900, 1911, 1919, 1923, 1935, 1940, 1952, 1960, 1966, 1968 (La Niña), 1986, 1993, 1998 (La Niña), and 2005 (El Niño) using split- and cross-validation (leave one out) approaches (Bennett et al, 2013). The Spearman Rho correlation among observed and reconstructed values was greater than 95% with a p-value of 0.001. The magnitude of climate variability over the past 2,000 years is visually apparent when inspecting plots for PDSI (Fig. 3.1 a, b, and c), temperature (Fig. 3.2 a, b,

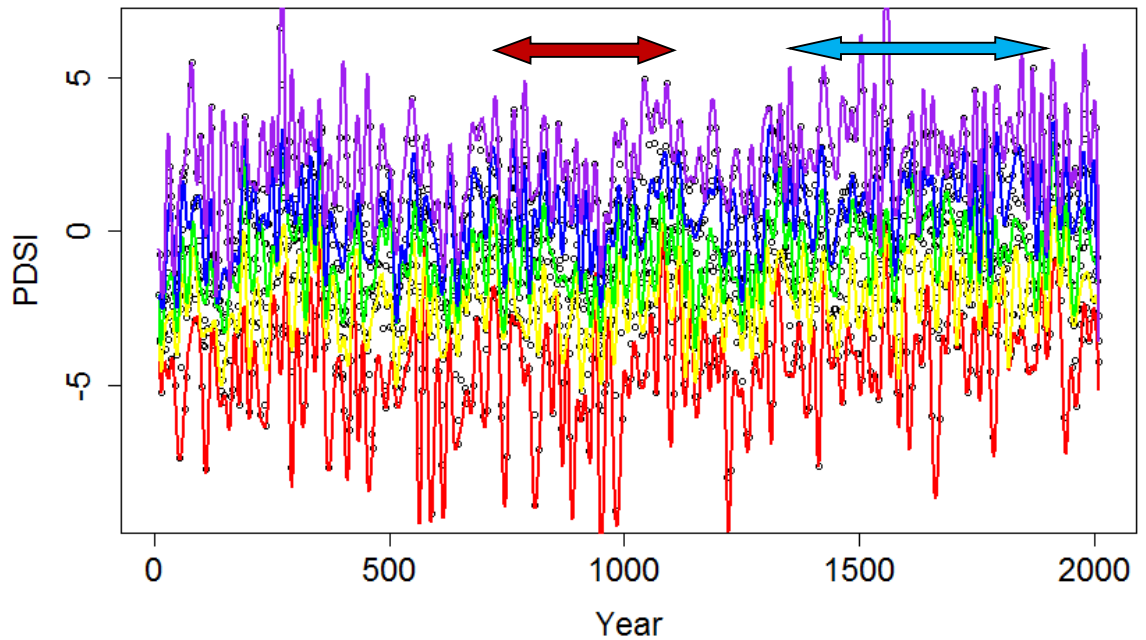
and c), and precipitation (Fig. 3.3 a, b, and c), where (a) is California, (b) is Colorado, and (c) is Nevada.



(a) California

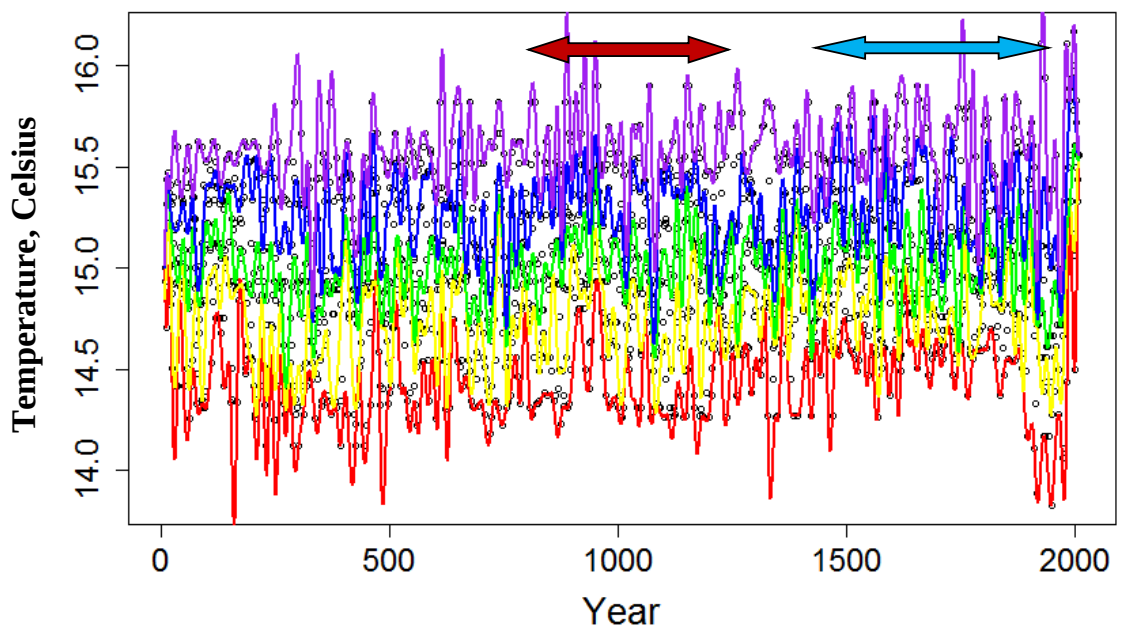


(b) Colorado

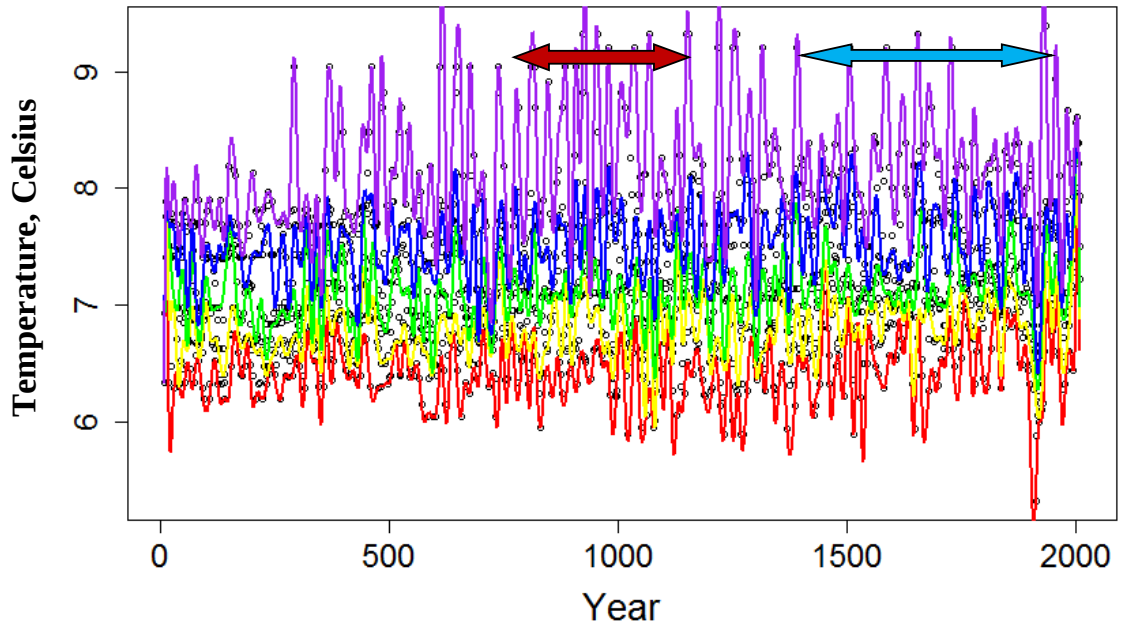


(c) Nevada

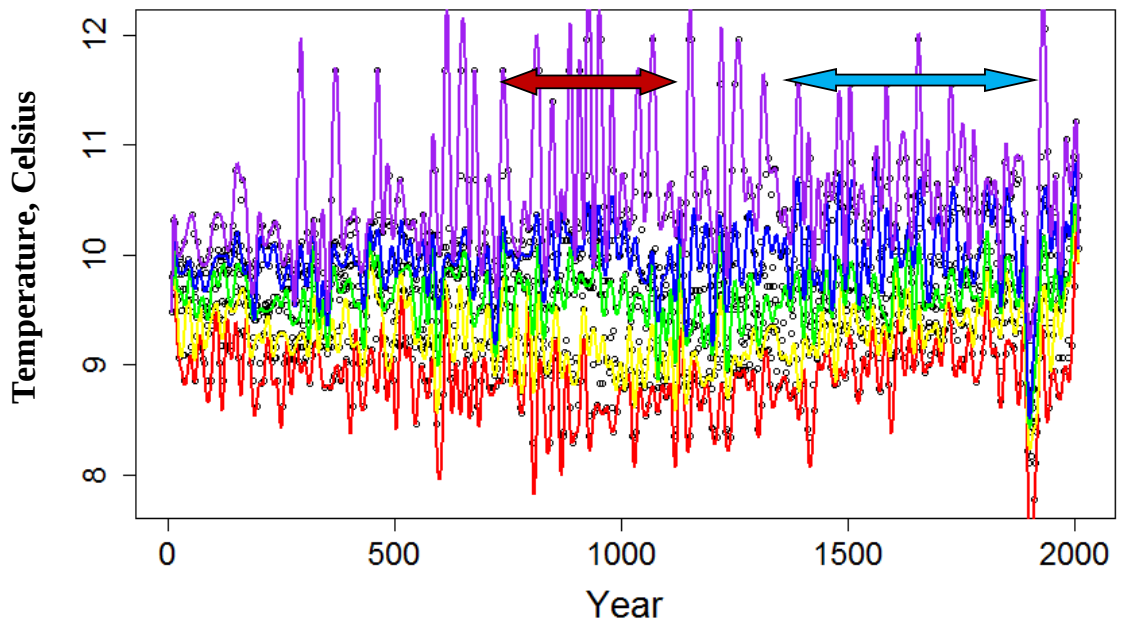
Fig. 3.1 - Quantile (upper panel) and histogram (lower panel) modeling of Palmer Drought Severity Index (PDSI) measurements (0-2009): (a) California, (b) Colorado, and (c) Nevada. The trends reflect 0.05 (red), 0.25 (yellow), 0.50 (green), 0.75 (blue), 0.95 (purple) quantiles. The approximate timing of the Medieval Warming Period (red) and Little Ice Age (blue) are indicated by double arrows.



(a) California

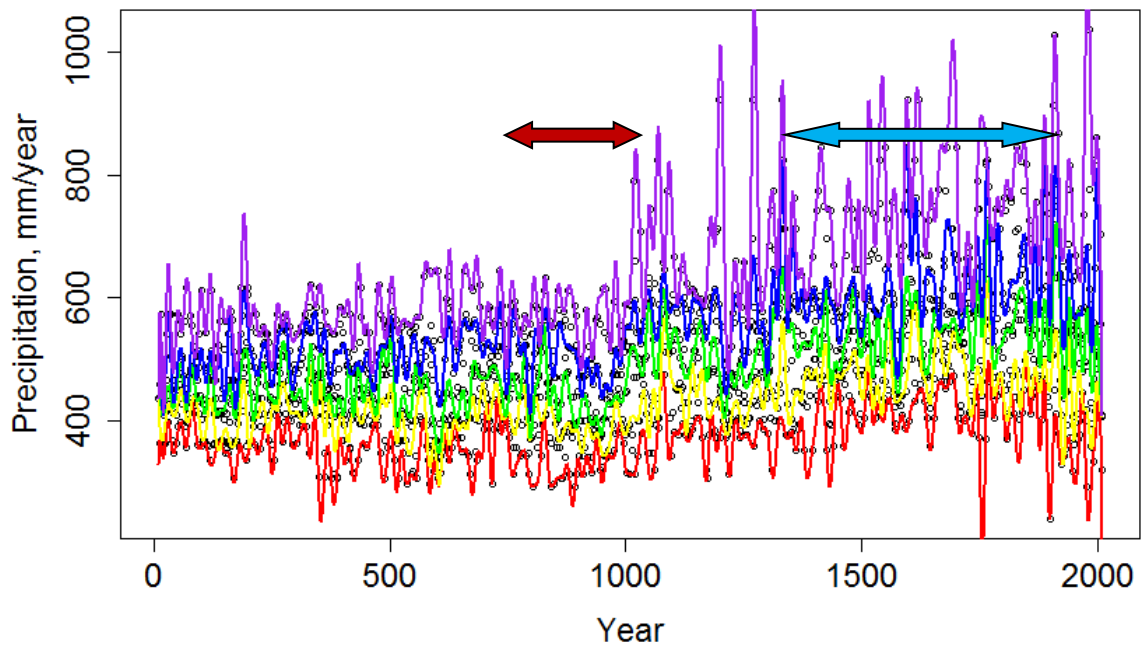


(b) Colorado

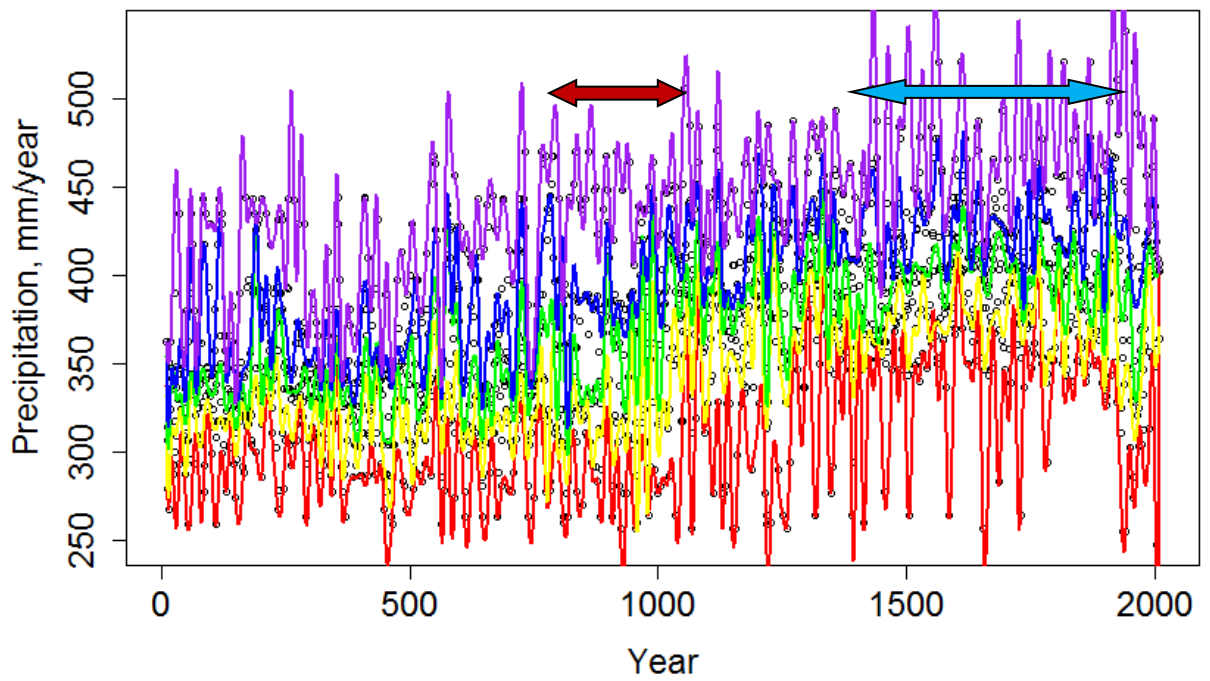


(c) Nevada

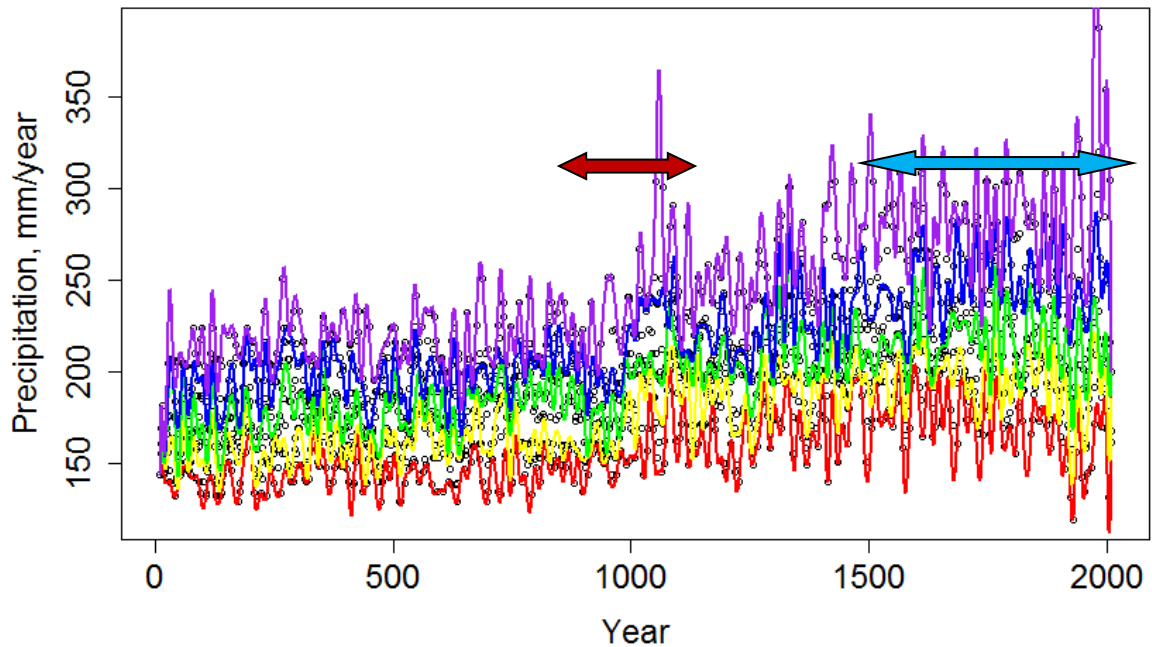
Fig. 3.2 - Quantile (upper panel) and histogram (lower panel) modeling of reconstructed temperature measurements (0-2009): (a) California, (b) Colorado, and (c) Nevada. The trends reflect 0.05 (red), 0.25 (yellow), 0.50 (green), 0.75 (blue), 0.95 (purple) quantiles. The approximate timing of the Medieval Warming Period (red) and Little Ice Age (blue) are indicated by double arrows.



(a) California



(b) Colorado



(c) Nevada

Fig. 3.3 - Quantile (upper panel) and histogram (lower panel) modeling of reconstructed precipitation measurements (0-2009): (a) California, (b) Colorado, and (c) Nevada. The trends reflect 0.05 (red), 0.25 (yellow), 0.50 (green), 0.75 (blue), 0.95 (purple) quantiles. The approximate timing of the Medieval Warming Period (red) and Little Ice Age (blue) are indicated by arrows.

In general, the plots reveal that all states in the southwest experienced past conditions ranging from extreme drought to extremely moist. It is interesting to note that over the past 2000 years, the ‘no-drought’ condition has the greatest frequency of occurrence in California (62.5%), Colorado (58.5%), and Nevada (55%). This suggests that while there is the likelihood for additional extreme events, there is a greater likelihood for current drought conditions to be eventually mitigated. Regarding temperature, the respective mode values in California, Colorado, and Nevada are about 14.8 °C (range from 14 °C to 16 °C), 6.75 °C (range from 5.5 °C to 9.5 °C), and 9.75 °C (range from 8 °C to 12 °C). Whereas the respective mode values for precipitation in California, Colorado, and Nevada are about 425 mm (range from 150 mm to 1050 mm),

350 mm (range from 230 mm to 540 mm), and 190 mm (range from 120 mm to 350 mm). Both the temperature and precipitation for these states reflect their association with the modern climate gradient.

3.2 Quantile trends in climate-change variables

Quantile regression modeling is applied to the past (0-2009) PDSI (Fig. 3.1 a, b, &c), temperature (Fig. 3.2 a, b, &c), and precipitation (Fig. 3.3 a, b, &c) data. The annual quantiles trends are presented for climate change data in a) California, b) Colorado, and c) Nevada. The decadal trends reflect 0.05 (red), 0.25 (yellow), 0.50 (green), 0.75 (blue), 0.95 (purple) quantiles determined using 200 b-spline degrees of freedom. In general, the quantile trends reveal that the long-term regional climate was interrupted by short-term changes. The so-called Medieval Warm Period (~900 to ~1250) and Little Ice Age (~1300 to ~1850) appear as primary lower frequency disruptions over the last two millennia in California and Nevada, and secondary (muted) disruptions in Colorado (Crowley and Lowery, 2000; Mann, 2002). The approximate timing of the Medieval Warming (horizontal red arrow) coincides with a decrease in PDSI (drier conditions), increase in temperature, and decrease in precipitation; whereas the Little Ice Age Periods (horizontal blue arrow) coincides with an increase in PDSI (wetter conditions), decrease in temperature, and increase in precipitation. These findings are attributed to strong ENSO teleconnections with California and Nevada, but mixed ENSO signals in Colorado (it is a region between El Niño and La Niña latitudes). In computing forecasts for future climate in southwestern states, additional quantile trends (0.05, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, and 0.95) are used to increase resolution when modeling the empirical cumulative distribution functions.

3.3 Forecasts of climate-change variables

3.3.1 A look at the past

To evaluate past climate-change data for long-memory process (Barbara, 1999), the PDSI, temperature, and precipitation are transformed to an equivalent fractal-information dimension. The transformation is applied to 20-year intervals resulting in time series plots comprising 100 lags. The various methods used to calculate the long-memory Hurst parameter (H) are summarized in table 3.1. Based on these H calculations, the precipitation appears to have the strongest long-memory process, whereas temperature and PDSI have a comparatively weaker long-memory process. This finding implies that the autocorrelation of precipitation is stronger than temperature; that is, the precipitation persists over a longer number of years than temperature.

Another finding is that the arithmetic calculation of H falls within the bounds of the Whittle approximation. One exception may be calculations based on the spectral regression method. For example, application of this method to precipitation results in an H parameter value that is outside the bounds of the Whittle approximation. This suggests that the spectral regression method might not provide reasonable approximations for comparatively strong long-memory processes. It is also interesting to note that the two driest states, Nevada and Arizona, have the lowest H values for precipitation. This suggests that wet cycles in these states are of shorter duration than other southwestern states. By contrast, Colorado has the second shortest temperature cycle and California the longest temperature cycle. At the same time, Colorado has the longest drought cycle and California has the shortest drought cycle. These findings suggest that Colorado is more likely to experience extended drought cycles at different temperatures, whereas

California is more likely to experience longer cycles of high temperatures that do not result in droughts, which can be explained by the fact that California borders the Pacific Ocean.

Table 3.1 – Summary of long-memory Hurst parameters for precipitation and temperature in Arizona, California, Colorado, New Mexico, Nevada, and Utah.

Precipitation					
	H ^a calculation		H interval Whittle Method		H via Spectral Regression
			2.50%	97.50%	
AZ	0.948		0.850	1.171	0.693
CA	0.978		0.869	1.234	0.701
CO	0.979		0.871	1.190	0.742
NM	0.983		0.930	1.368	0.742
NV	0.964		0.832	1.152	0.693
UT	0.982		0.934	1.312	0.999
Temperature					
	H ^a calculation		H interval Whittle Method		H via Spectral Regression
			2.50%	97.50%	
AZ	0.606		0.569	0.651	0.616
CA	0.662		0.546	0.790	0.597
CO	0.579		0.533	0.628	0.592
NM	0.602		0.521	0.738	0.600
NV	0.560		0.521	0.611	0.570
UT	0.605		0.647	0.895	0.601
Palmer Drought Severity Index					
	H ^a calculation		H interval Whittle Method		H via Spectral Regression
			2.50%	97.50%	
AZ	0.639		0.590	0.696	0.647
CA	0.554		0.507	0.604	0.538
CO	0.651		0.596	0.707	0.682
NM	0.633		0.574	0.692	0.649
NV	0.630		0.572	0.688	0.646
UT	0.598		0.533	0.664	0.638

3.3.2 A look at the present

The previous section established that the southwest climate-change reconstructions are stationary and stochastically self-similar. Given these facts, the

FARIMA model is applied to individual quantile trends for forecasting of climate-change variables. In all cases, the fitted FARIMA models are characterized as one of the following types: from AR(0) MA(0) to AR(0) MA(6). Using the fitted FARIMA models, forecasts are generated over the range of quantiles from which ECDFs are then computed for each variable.

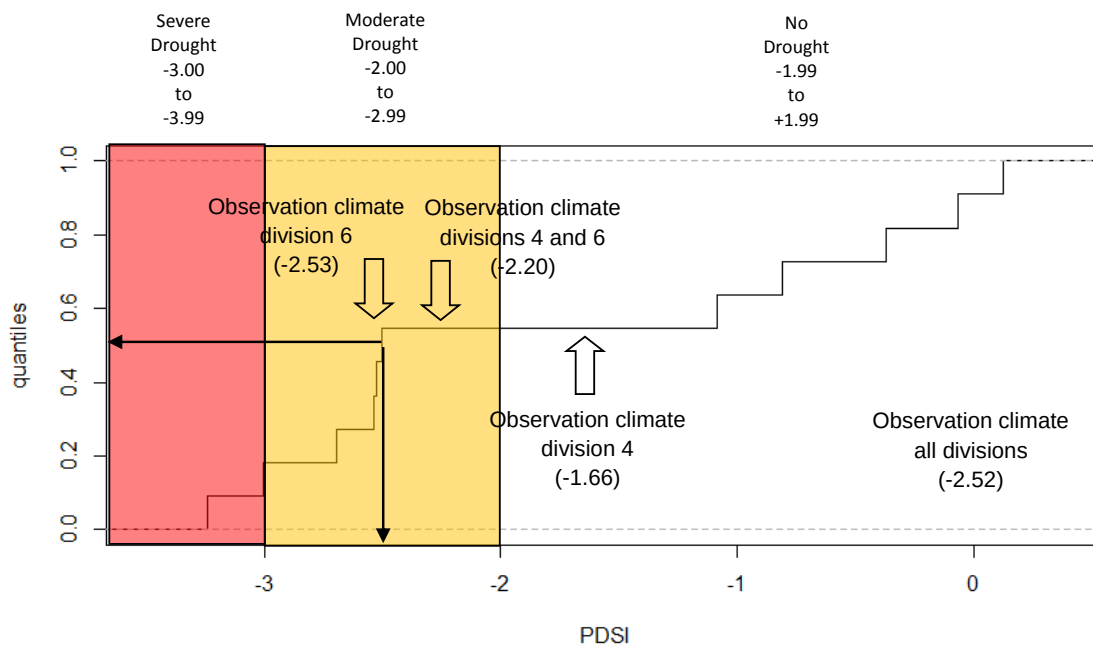
The FARIMA model performance is evaluated by comparing the median 2012 state forecasts of PDSI (Fig. 3.4), temperature (Fig. 3.5), and precipitation (Fig. 3.6) (California, Colorado, and Nevada respectively) to state observations (National Oceanic and Atmospheric Administration, 2013). In this comparison, the PDSI reflects values averaged over the period of June-July-August, the temperature represents values averaged over the period of January-December, and the precipitation reflect values accumulated precipitation over the period of January-December. When comparing the 2012 climate-change observations to forecasts, it is important to note the differences in their spatiotemporal representation. Specifically, the observed PDSI and reconstructed climate variables are values associated with a grid location (influenced by a 250 km x 250 km region) defined by Cook et al. (2004), whereas the observed climate variables represent values averaged over climate divisions with larger areas defined by the National Oceanic and Atmospheric Administration (2013).

The respective 2012 state observations and forecasts of PDSI, temperature, and precipitation are presented in Figs. 3.4-3.6. Reviewing the plots for California, Colorado, and Nevada reveals that all observations plot within the probable forecast limits supporting the usefulness of the hybrid approach. In addition, the range of probable climate variable differs among states reflecting the modern climate gradient and

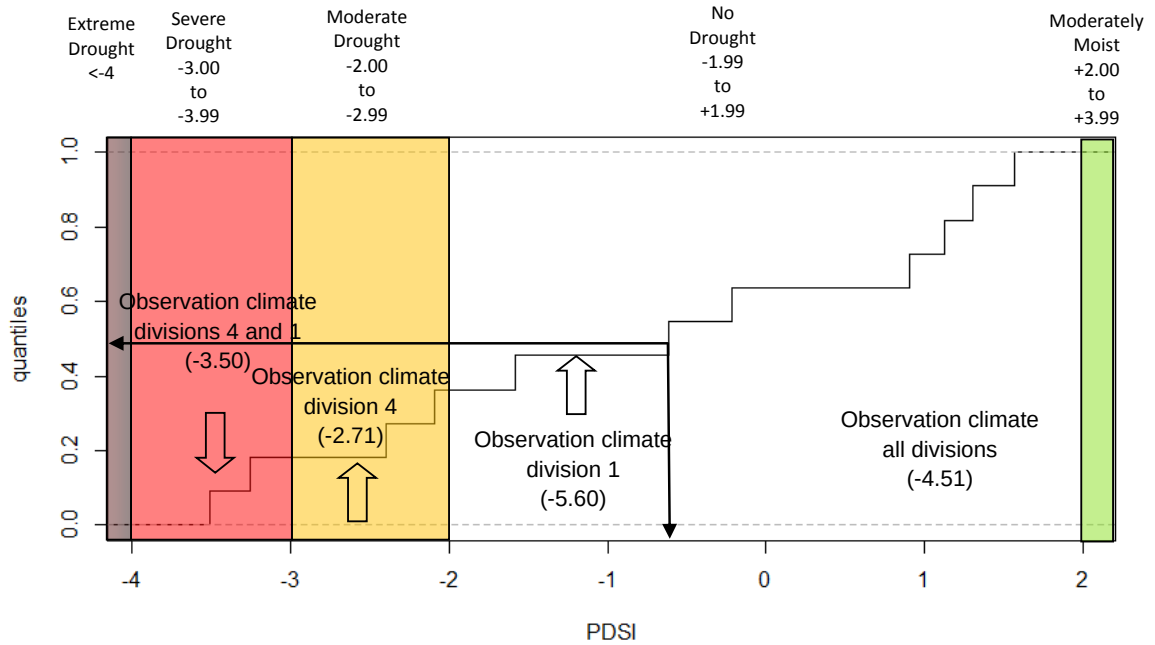
underscoring the stationarity of conditions over the past 2000 years. For example, the respective forecast range of PDSI (Fig. 3.4) for California, Colorado, and Nevada is from -3.50 to 0.5, from -4.25 to 2.25, and from -5.0 to 1.0. The respective forecast range of temperature (Fig. 3.5) for California, Colorado, and Nevada is from 15.1 °C to 15.8 °C, from 6.65 °C to 8.0 °C, and from 9.6 °C to 10.5 °C. The respective forecast range of precipitation (Fig. 3.6) for California, Colorado, and Nevada is from 200 mm to 625 mm, from 340 mm to 440 mm, and from 170 mm to 260 mm.

Comparing the median forecast (indicated by vertical arrows associated with the 0.5 quantile) to observed climate variables averaged over different climate divisions reveals reasonable correspondence. For example, the respective median forecast and averaged observations of PDSI in California (Fig. 3.4a) are about -2.50 and -2.20 (climate divisions 4 and 6). The respective median forecast and averaged observation of PDSI in Colorado (Fig. 3.4b) are about -0.75 and -2.71 (climate divisions 4 and 1); and the respective median forecast and averaged observation of PDSI in Nevada (Fig. 3.4c) are about -2.15 and -2.17 (climate divisions 3 and 4). Similarly, the respective median forecast and averaged observation of temperature in California (Fig. 3.5a) are about 15.44 °C and 15.2 °C (climate divisions 4 and 6). The respective median forecast and averaged observation of temperature in Colorado (Fig. 3.5b) are about 7.42 °C and 7.78 °C (climate divisions 4 and 1); and the respective median forecast and averaged observation of temperature in Nevada (Fig. 3.5c) are about 10.05 °C and 10.6 °C (climate divisions 3 and 4). The respective median forecast and averaged observation of precipitation in California (Fig. 3.6a) are about 450 mm and 440 mm (climate divisions 4 and 6). The respective median forecast and averaged observation of precipitation in Colorado (Fig. 3.6b) are

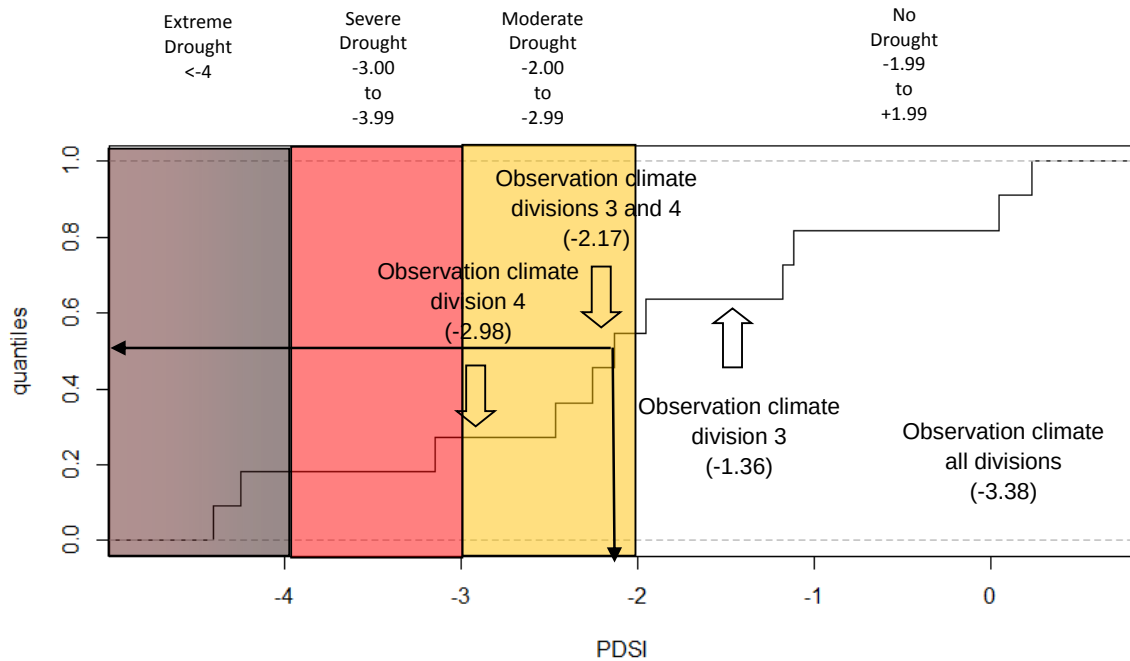
about 403 mm and 392 mm (climate divisions 4 and 1); and the respective median forecast and averaged observation of precipitation in Nevada (Fig. 3.6c) are about 202 mm and 188.5 mm (climate divisions 3 and 4). Comparison of median forecasts to values for individual divisions and all state divisions reveals the heterogeneous nature of climate variables within each state; that is, the observations across climate regions in California are mostly Mediterranean; in Colorado, they range from semiarid to alpine; in Nevada they range from semiarid to arid. This finding suggests that future improvements, potentially spanning five to 15 years, to forecasting may be achieved by introducing PDSI data associated with grid nodes across all state climate divisions.



(a) California

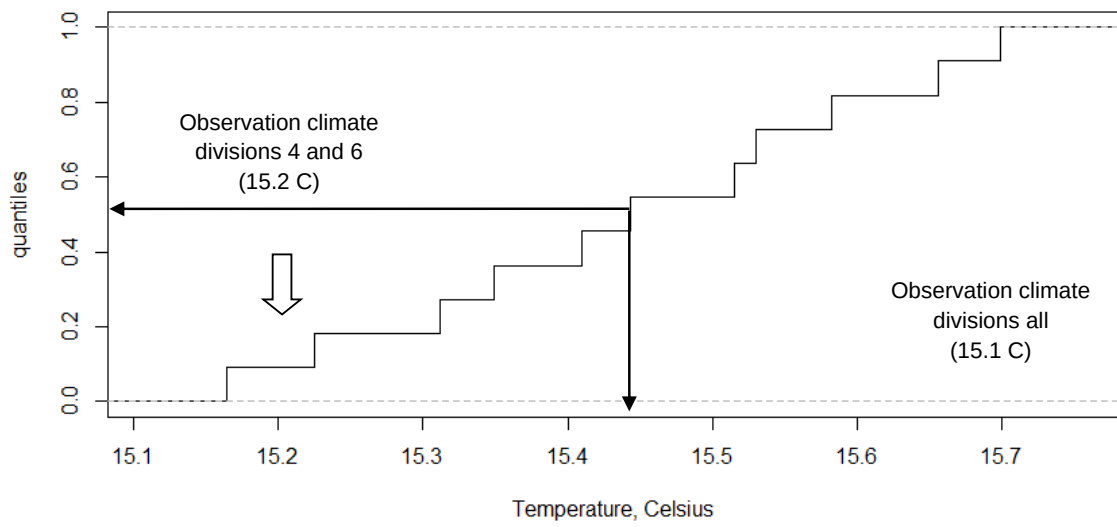


(b) Colorado

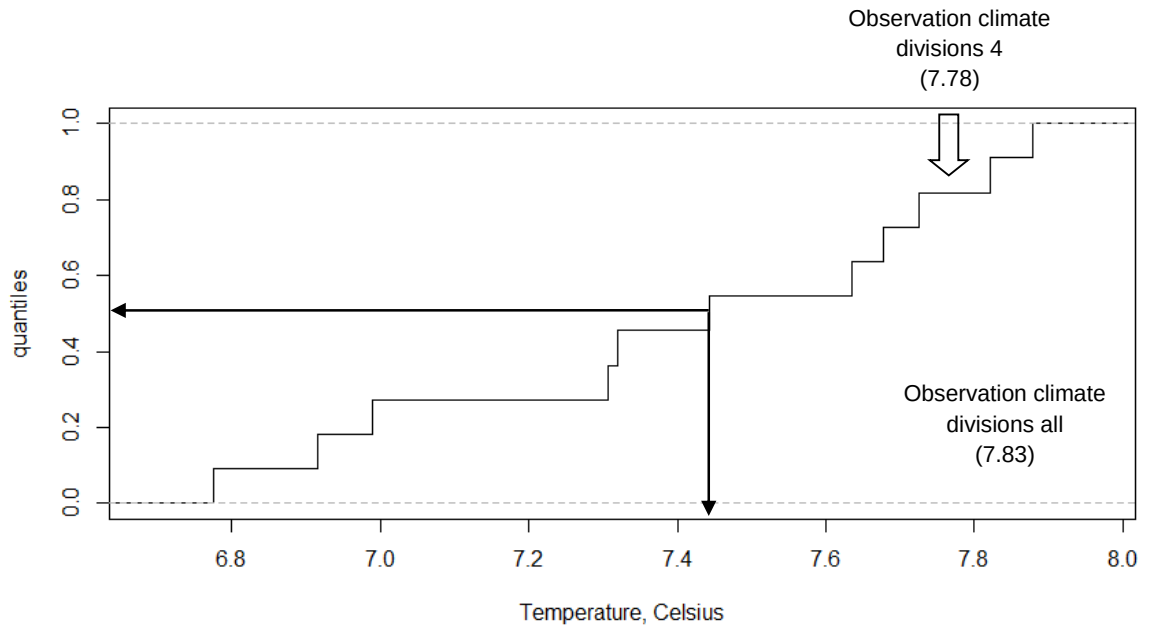


(c) Nevada

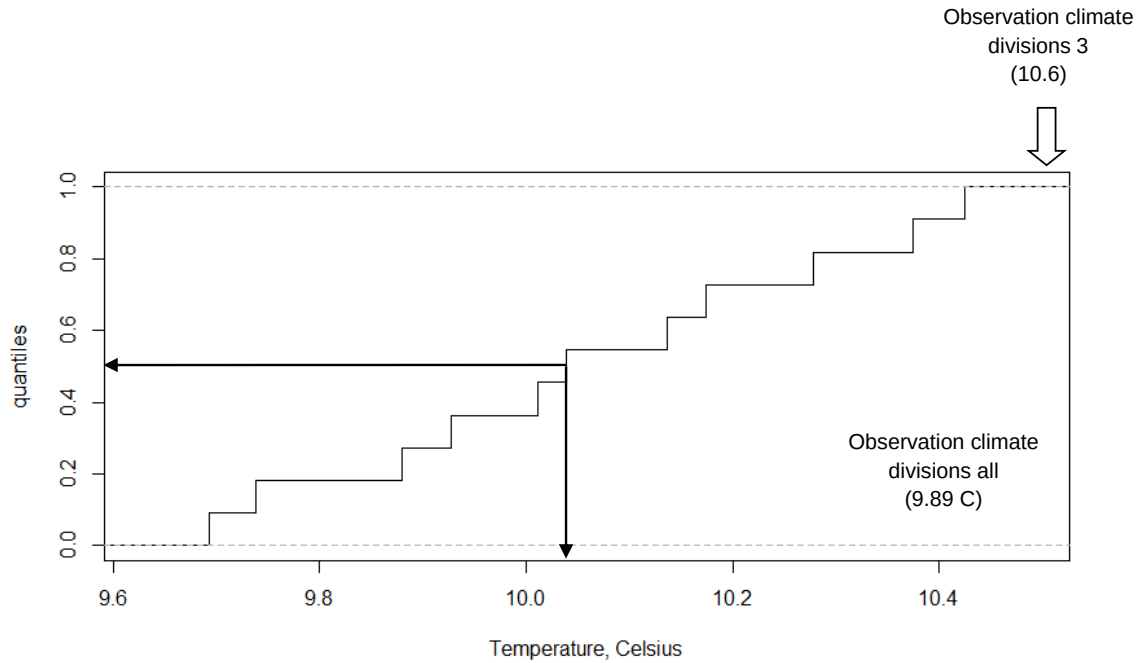
Fig. 3.4 - Comparison of conditional forecasts and measurement observations for 2012 Palmer Drought Severity index (PDSI): (a) California, (b) Colorado, and (c) Nevada. Arrows denote median forecasted (thin) and observed (open) PDSI value.



(a) California

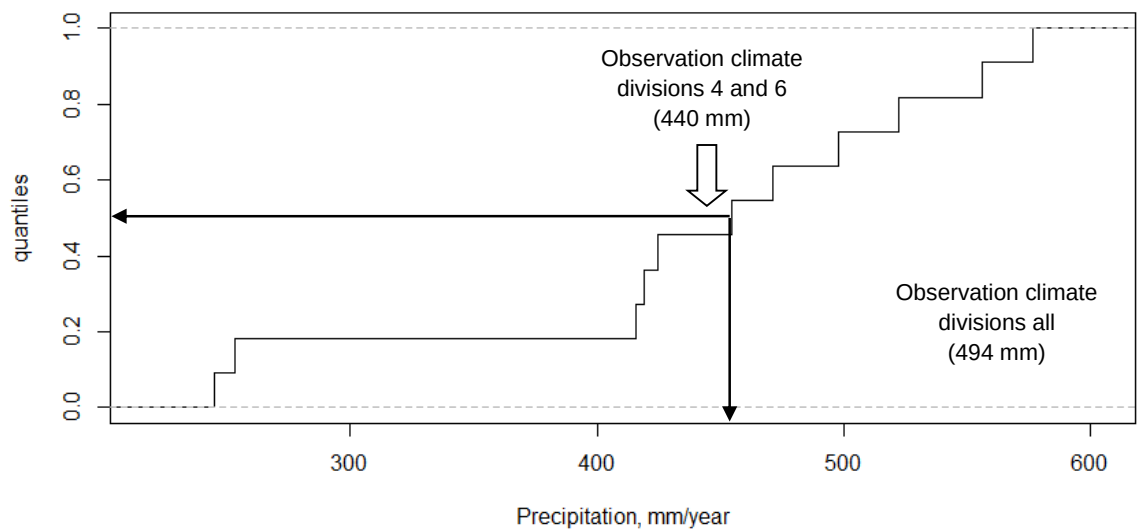


(b) Colorado

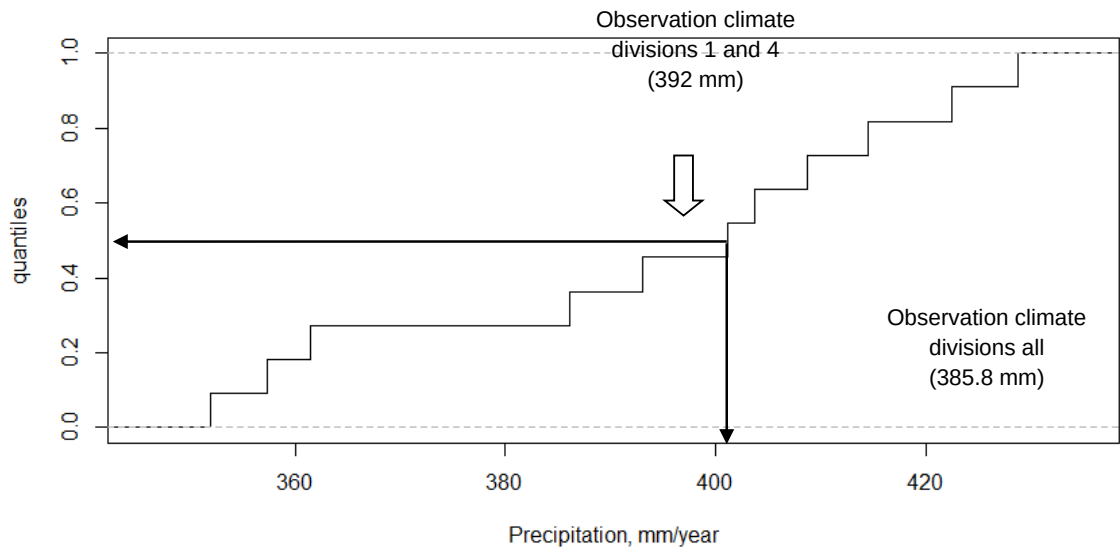


(c) Nevada

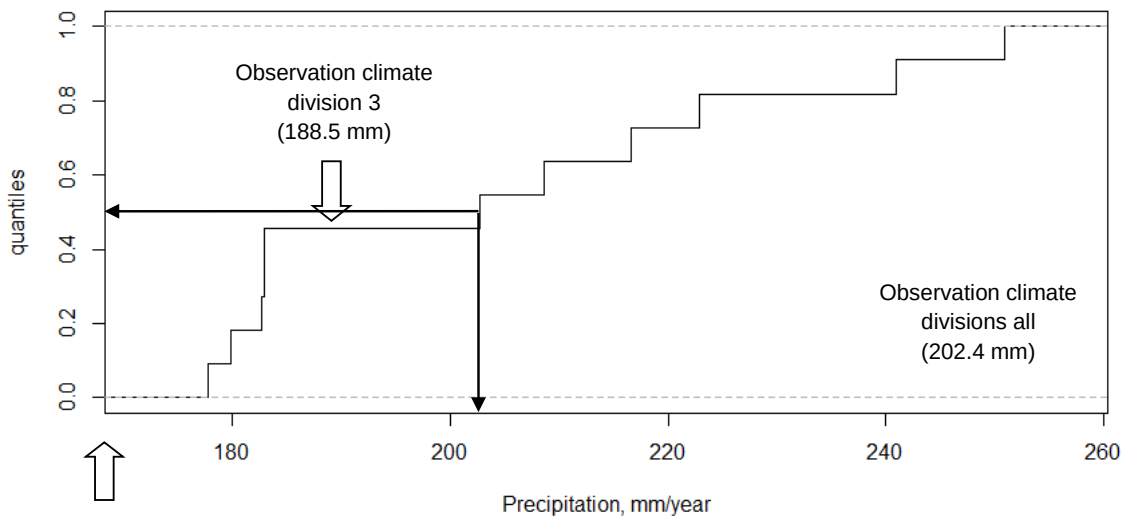
Fig. 3.5 - Comparison of conditional forecasts and measurement observations for 2012 temperature: (a) California, (b) Colorado, and (c) Nevada. Arrows denote median forecasted (thin) and observed (open) temperature value.



(a) California



(b) Colorado



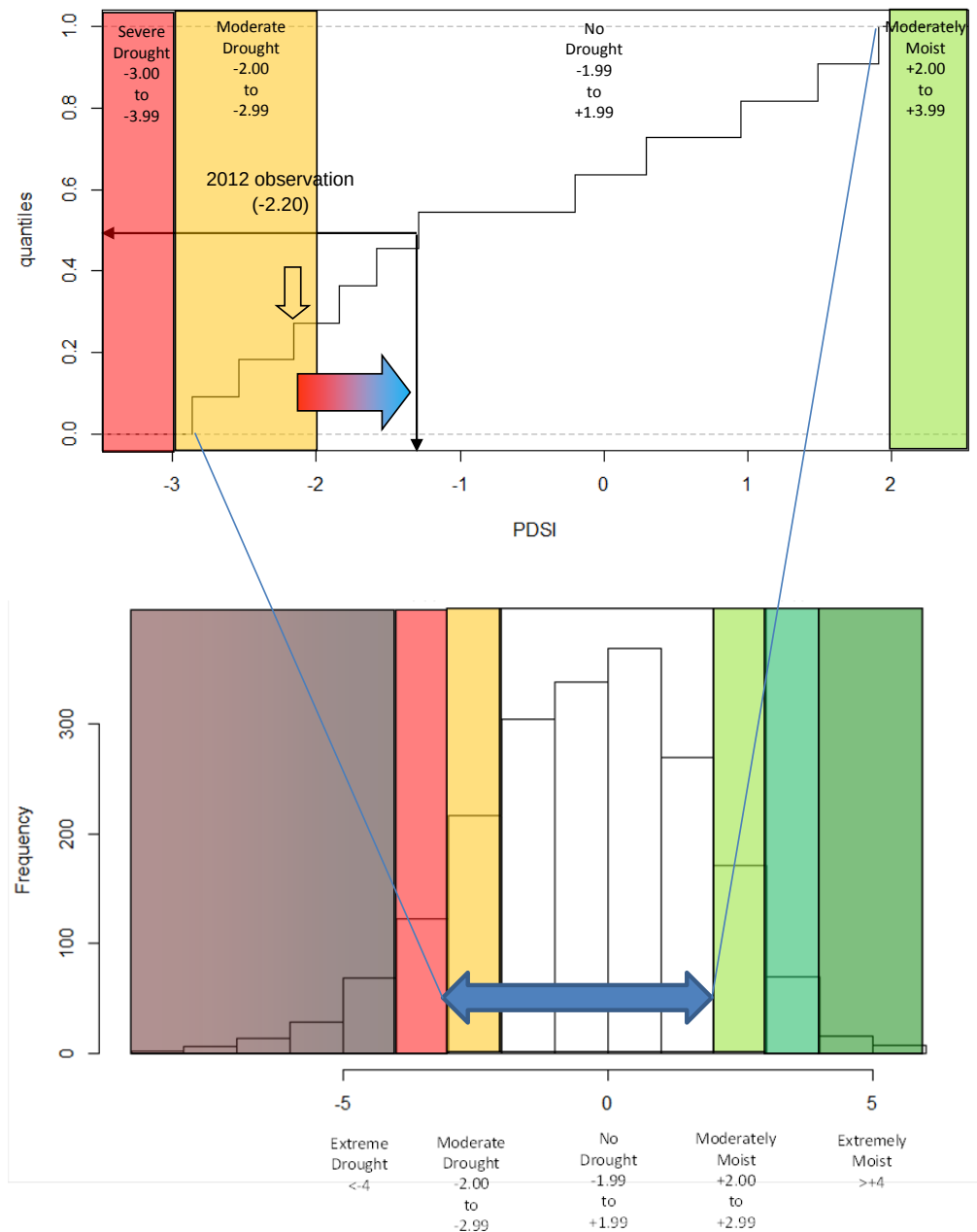
Observation climate divisions 3 and 4 (147 mm)

(c) Nevada

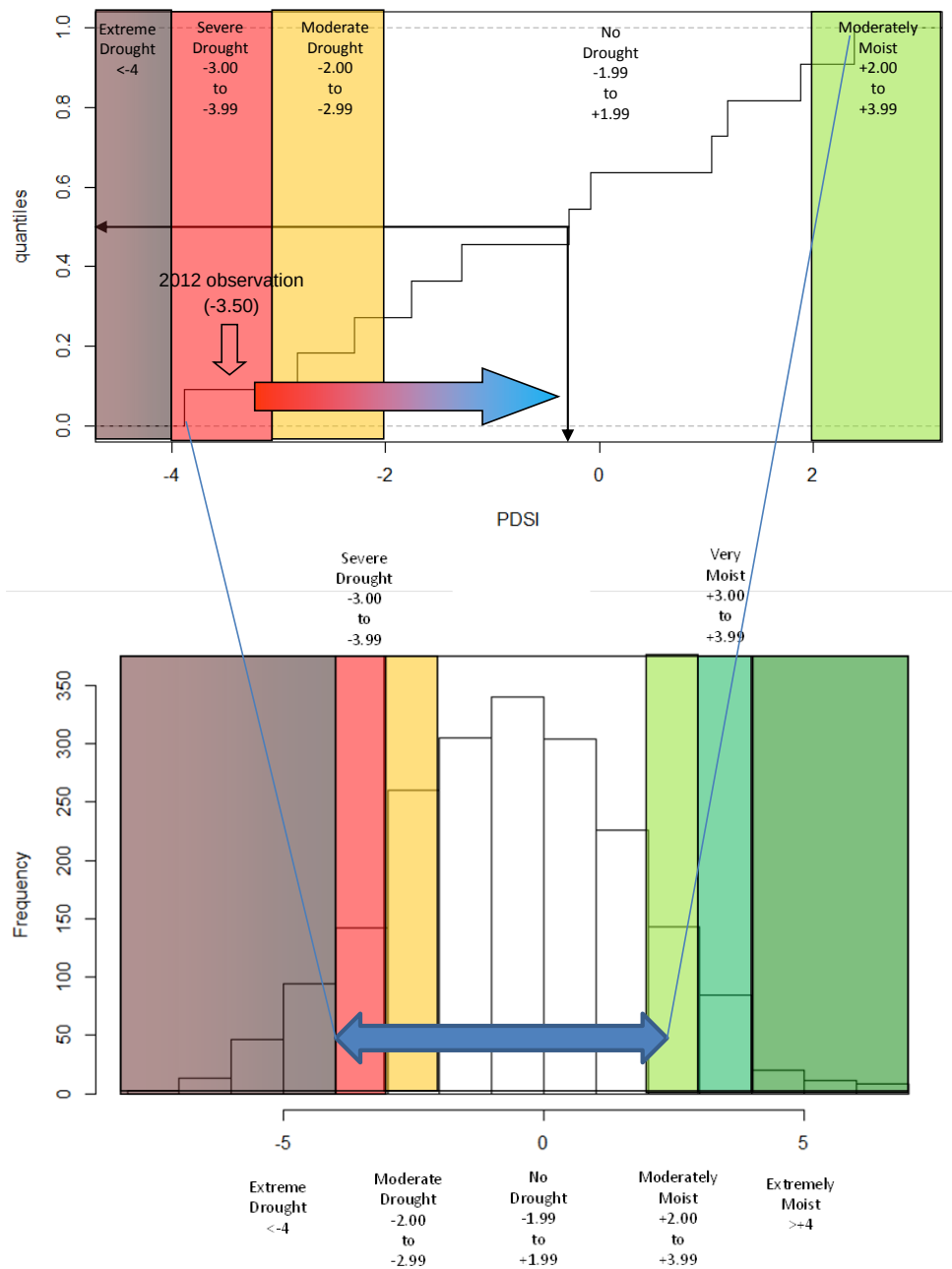
Fig. 3.6 - Comparison of conditional forecasts and measurement observations for 2012 precipitation: (a) California, (b) Colorado, and (c) Nevada. Arrows denote median forecasted (thin) and observed (open) precipitation value.

3.3.2 A look at the future

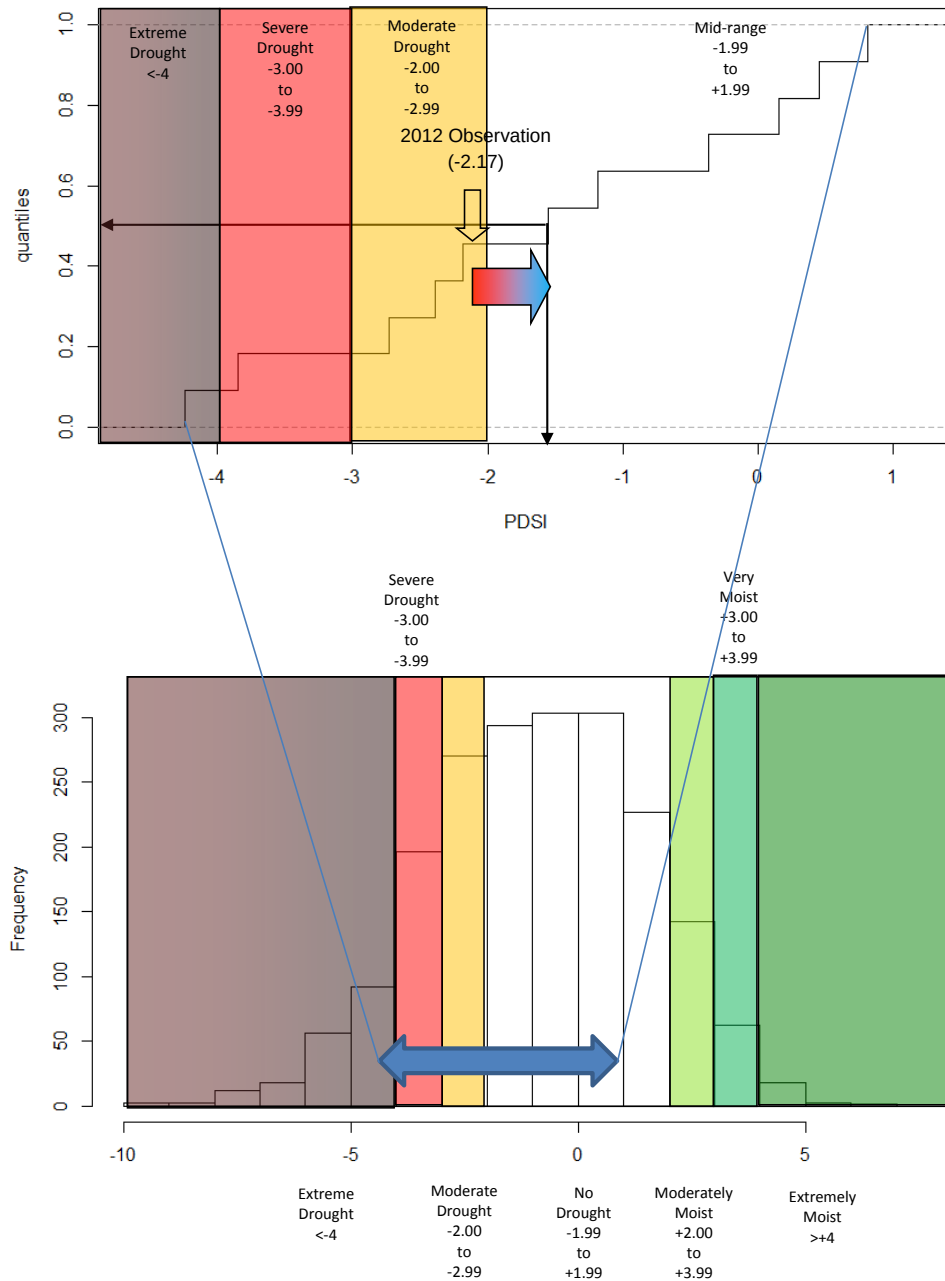
The conditional state 2030 forecasts are presented together with 2012 observations (average of climate divisions) for PDSI, temperature, and precipitation (Figs. 3.7-3.9). For example, the conditional forecasts of PDSI in California, Colorado, and Nevada are presented in Fig. 3.7.



(a) California



(b) Colorado

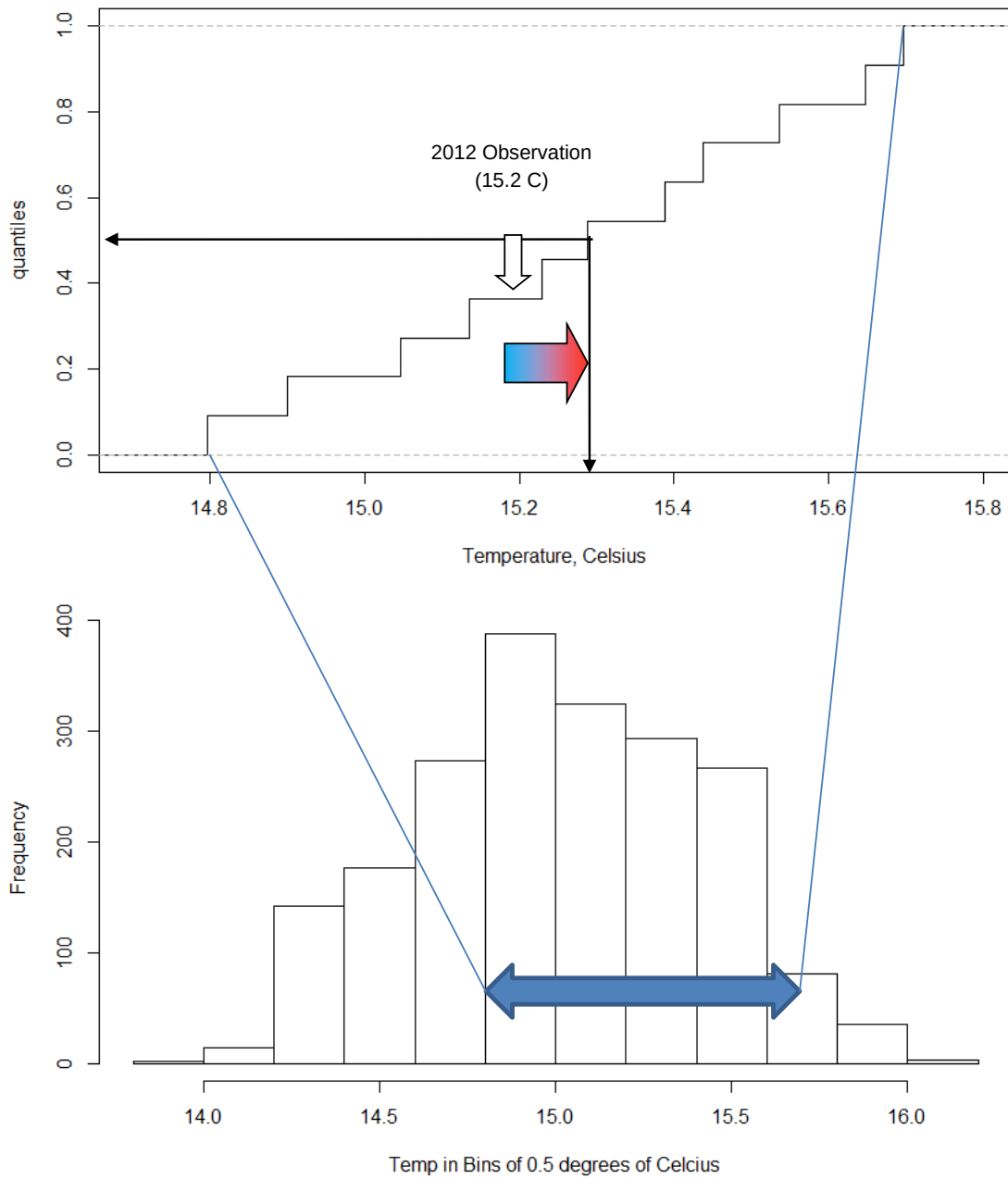


(c) Nevada

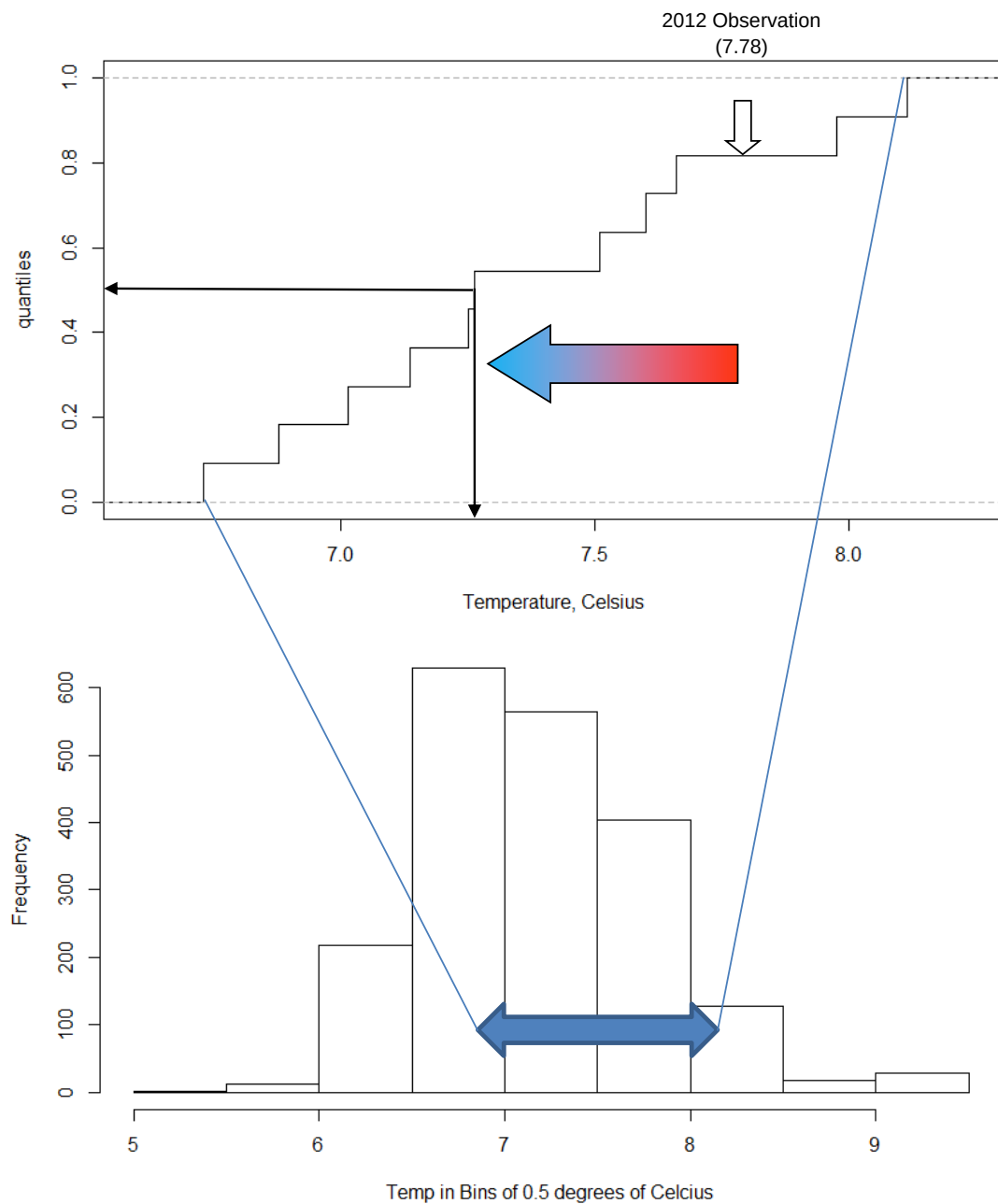
Fig. 3.7 - Conditional forecasts and histogram (lower panel) for Palmer Drought Severity Index in the year 2030: (a) California, (b) Colorado, and (c) Nevada. Arrows denote median forecasted 2030 (thin) and observed 2012 (open) Palmer Drought Severity Index values.

In California (Fig. 3.7a), the median PDSI forecast value of about -1.42 is larger than the observed 2012 value of about -2.20 (average of climate divisions 4 and 6). This

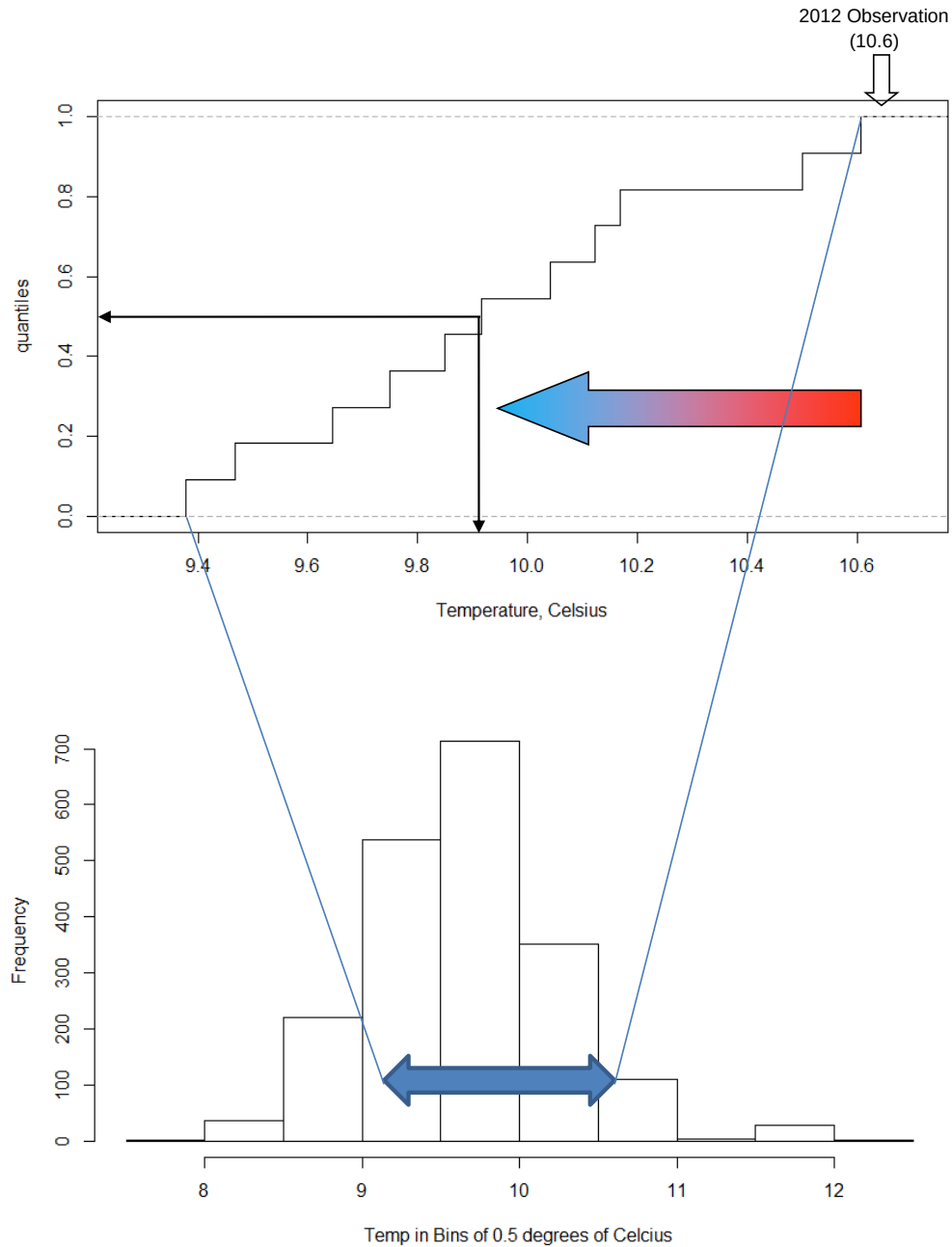
suggests the likelihood for a shift toward moister conditions; that is, the current moderate drought condition is likely to shift to a mid-range condition within the next 8 years. In Colorado (Fig. 3.7b), the median PDSI forecast value of about -0.40 is larger than the observed 2012 value (average of climate divisions 4 and 1) of about -3.50, also suggesting a shift from severe drought to a mid-range condition in the next 8 years. In Nevada (Fig. 3.7c), the median PDSI forecast value of about -1.60 is larger than the observed 2012 value of about -2.17 (average of climate divisions 4 and 6), also suggesting the likelihood for a shift toward moister conditions in the next 8 years. In addition to trends based on median values, there also is some probability that California could experience conditions ranging from severe drought to moderately moist in the next 8 years. Colorado could experience conditions ranging from extreme drought to moderately moist in the next 8 years and Nevada could experience conditions ranging from severe drought to mid-range in the next 8 years. Relative to Colorado, California tends toward the moist and Nevada shifts toward drought conditions.



(a) California



(b) Colorado

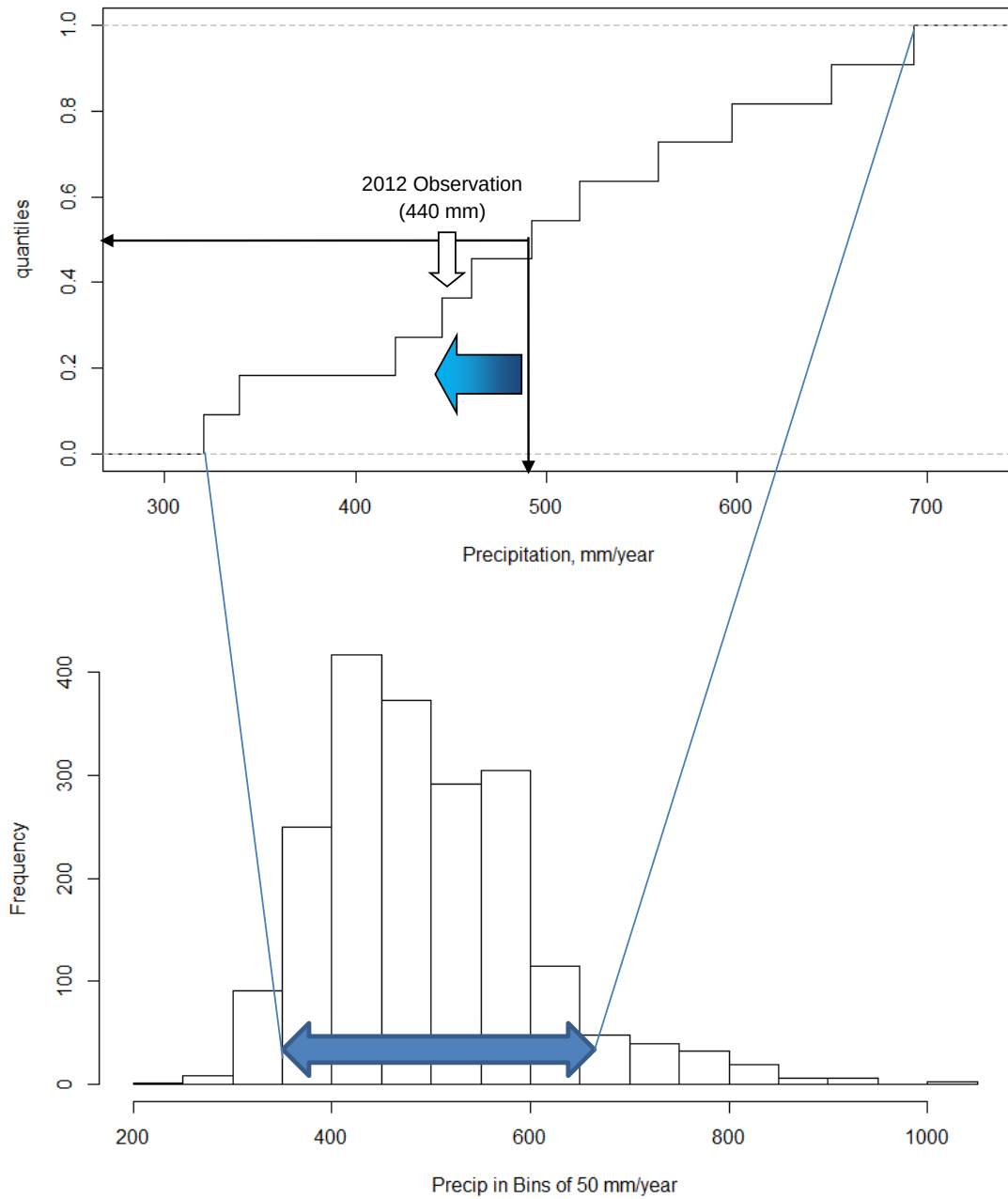


(c) Nevada

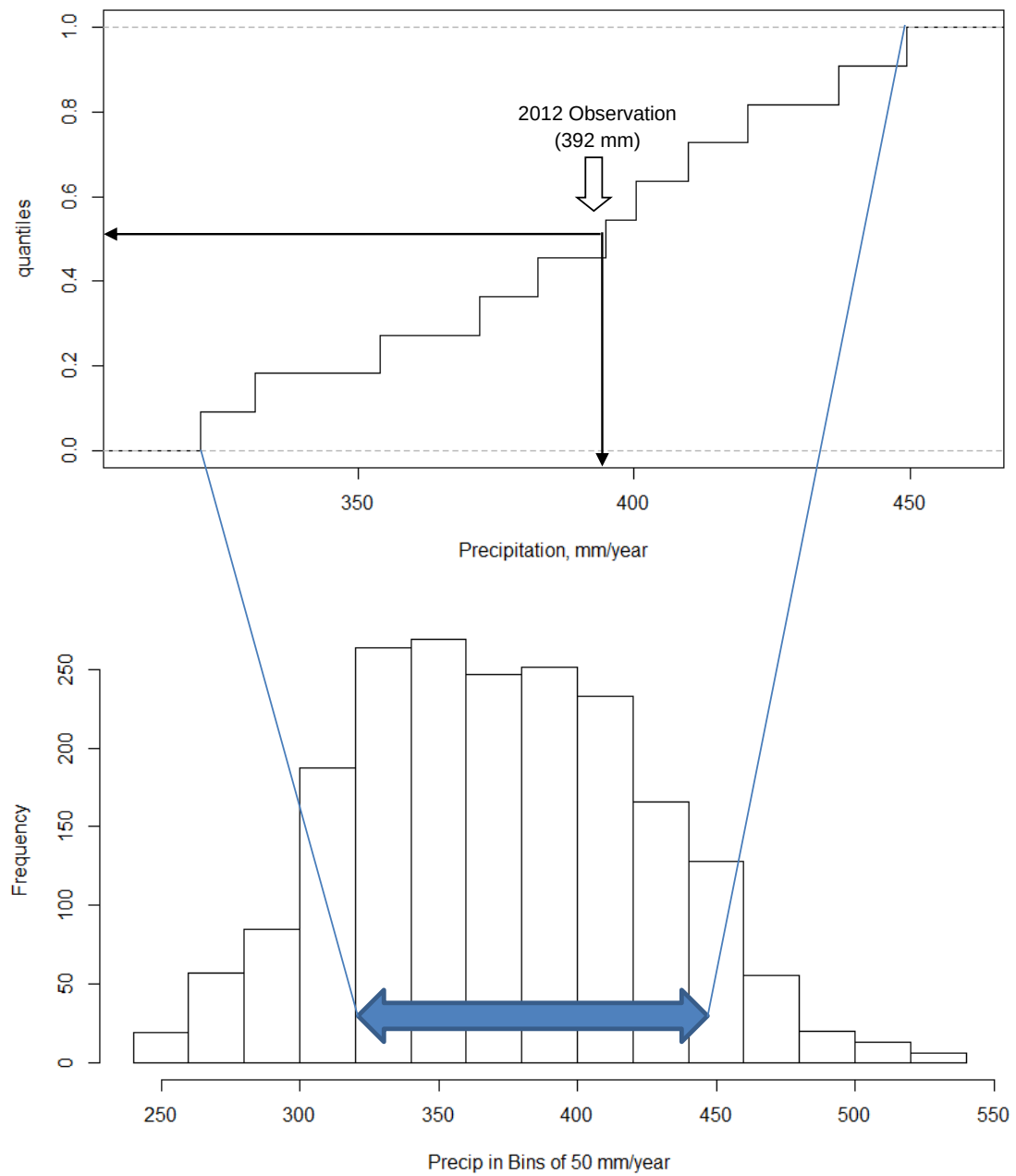
Fig. 3.8 - Conditional forecasts and histogram (lower panel) for temperature in the year 2030: (a) California, (b) Colorado, and (c) Nevada. Arrows denote median forecasted 2030 (thin) and observed 2012 (open) temperature values.

The conditional state 2030 forecasts of temperature in California, Colorado, and Nevada are presented in Fig. 3.8. In California (Fig. 3.8a), the median temperature

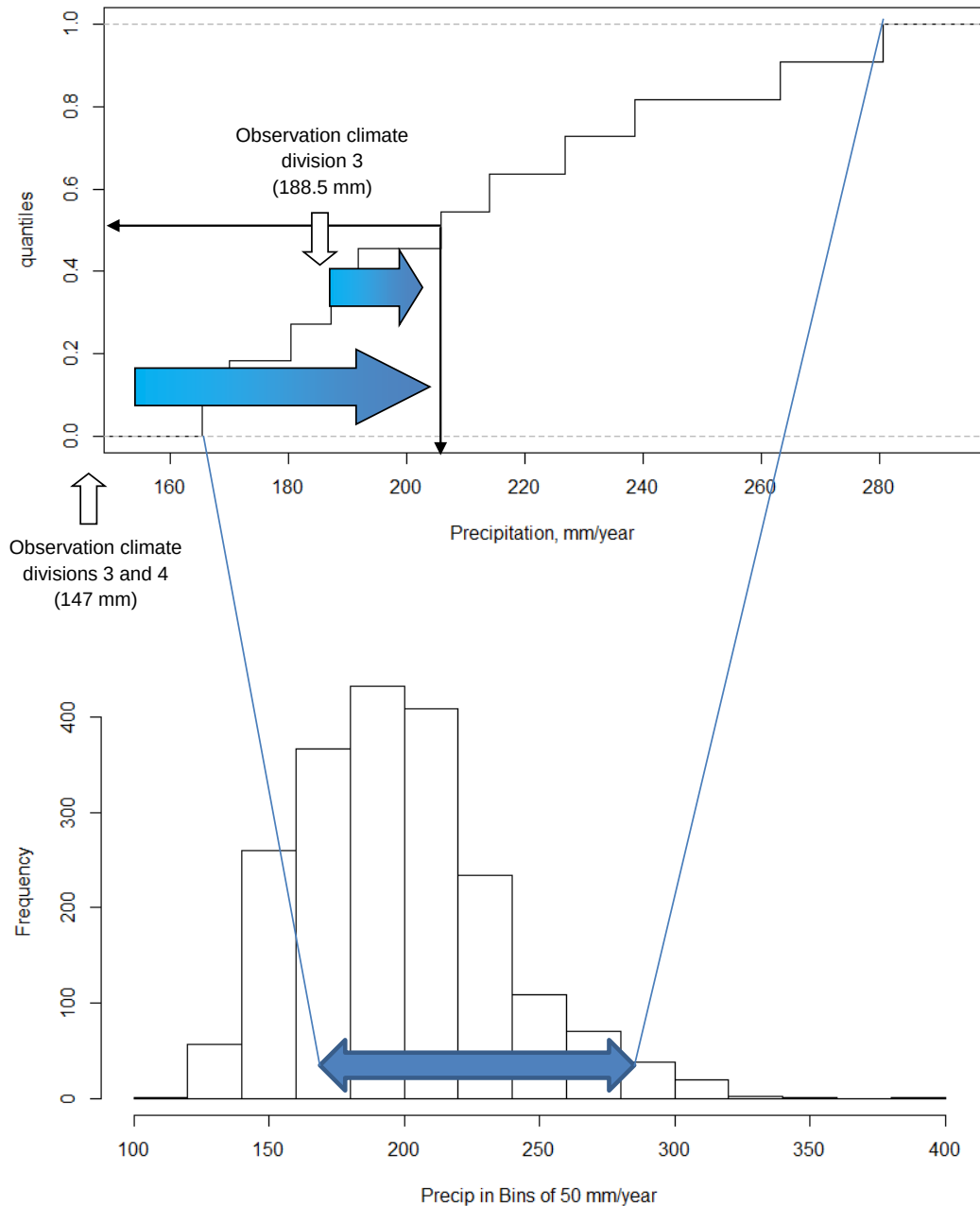
forecast value of about 15.28 °C is about the same as the observed 2012 temperature (average of climate divisions 4 and 6) of about 15.2 °C. This suggests the likelihood for a slight shift toward warmer conditions in the next few years. In Colorado (Fig. 3.8b), the median temperature forecast value of about 7.3 °C is less than the 2012 temperature (average of climate divisions 4) of about 7.78 °C, suggesting a shift toward cooler conditions in the next few years. In Nevada (Fig. 3.8c), the median temperature forecast value of about 9.9 °C is smaller than the 2012 temperature (average of climate divisions 4 and 6) of about 10.6 °C, suggesting the likelihood for cooler conditions in the next few years. In addition to trends based on median values, there also is some probability that California could experience temperatures ranging from about 14.7 °C to 15.8 °C, Colorado could experience temperatures ranging from about 6.5 °C to 8.25 °C, and Nevada could experience temperatures ranging from about 9.2 °C to 10.7 °C. Within the next two to eight years, relative to Colorado, California and Nevada tend toward warmer conditions.



(a) California



(b) Colorado



(a) Nevada

Fig. 3.9 - Conditional forecasts and histogram (lower panel) for precipitation in the year 2030: (a) California, (b) Colorado, and (c) Nevada. Arrows denote median forecasted 2030 (thin) and observed 2012 (open) precipitation values.

The conditional state 2030 forecasts of precipitation in California, Colorado, and Nevada are presented in Fig. 3.9. In California (Fig. 3.9a), the median precipitation

forecast value of about 490 mm is larger than the 2012 precipitation value (average of climate divisions 4 and 6) of about 440 mm. This suggests the likelihood for a future shift toward drier conditions in the next few years. In Colorado (Fig. 3.9b), the median precipitation forecast value of about 390 mm is about the same as the 2012 precipitation value (average of climate divisions 4) of about 392 mm, suggesting no change in precipitation conditions. In Nevada (Fig. 3.9c), the median temperature forecast value of about 205 mm is larger than the historical median value of about 188.5 mm, suggesting the likelihood for future wetter conditions in roughly eight years. In addition to trends based on median values, there also is some probability that California could experience precipitation amounts ranging from 275 mm to 725 mm, Colorado could experience precipitation amounts ranging from 300 mm to 470 mm, and Nevada could experience precipitation amounts ranging from 150 mm to 300 mm within the next eight years. Relative to Colorado, California tends toward wetter conditions and Nevada toward drier conditions.

In addition to evaluating forecasts for the year 2030, an analysis of trends is conducted over forecasts spanning a 50-year period (from 2010 to 2060). A summary of findings is presented for the southwestern states in Table 2. These results indicate that the southwestern US is likely to experience decreasing temperatures that will likely continue over the next 25 years resulting in wetter and cooler conditions for the region. The decay represents a decrease of 0.20 degrees Celsius over 20 years in temperature. The accuracy of the forecast results for temperatures are verified by comparing the root mean squared error (RMSE) (Bennett et al, 2013; Hyndman and Koehler, 2006) of the forecast of each

state to the mean of the forecast of that state. For temperature, the range is from 2.3% to 8.2% of RMSE to the mean indicating a low variability.

The PDSI index will increase in the same period, indicating wetter conditions across the southwestern states. However, this increase is only by about 0.8 on the PDSI index, which suggests only a very moderate decrease of drought. The exception to this increase is California where the forecast suggests a constant level of PDSI suggesting that California has reached its equilibrium, the end of a long-memory process. Since PDSI values are mainly negative, RMSE is not a good indicator of performance, since it relies on the absolute values, instead the mean error (ME) of the residuals is chosen here. A comparison between the ME of each state and the mean of the forecast of each state indicates variability of about 2% to 4%.

Table 3.2 - Forecast trends (2000-2050) for temperature, precipitation, and drought in Arizona, California, Colorado, New Mexico, Nevada, and Utah. Results are for 0.25, 0.50, and 0.75 quantiles.

Palmer Drought Severity Index (PDSI)				
	Forecast Accuracy		Mean of Series	Forecast Direction
State	ME ¹			
AZ	0.020		-0.54	increase*
CA	0.016		-0.24	constant
CO	0.011		-0.51	increase*
NM	0.014		-0.61	increase*
NV	0.018		-0.70	increase*
UT	0.009		-0.45	increase*
Increase = increase of 0.8 to 1.0 on PDSI scale over 10 years Decrease = decrease of 0.8 to 1.0 on PDSI scale over 10 years * Since PDSI is measured from positive to negative, increase indicates wetter conditions				
Precipitation				
	Forecast Accuracy		Mean of Series	Forecast Direction
State	RMSE ²	MAE ³		
AZ	59.5	44.2	306.1	sharp increase
CA	98.3	77.0	528.2	increase
CO	45.7	35.7	387.7	decrease
NM	55.6	41.9	333.8	increase
NV	31.3	24.9	215.0	decrease
UT	39.4	31.1	283.9	increase
Increase = increase of 5 mm over 10 years Sharp Increase = increase of 20 mm over 10 years Decrease = decrease of 5 mm over 10 years Sharp Decrease = decrease of 20 mm over 10 years				
Temperature				
Median of Series	Forecast Accuracy		Mean of Series	Forecast Direction
	RMSE	MAE		
AZ	0.62	0.51	15.6	decrease
CA	0.39	0.32	15.1	decrease
CO	0.59	0.47	7.2	decrease
NM	0.46	0.37	11.8	decrease
NV	0.58	0.45	9.7	decrease
UT	0.60	0.50	9.0	decrease
Decrease = decrease of 0.20 degrees over 20 years Increase = increase of 0.20 degrees over 20 years				

¹ ME – mean error, since PDSI is mostly a negative value, MAE and RMSE do not apply here.

² RMSE – root mean squared error.

³ MAE – mean absolute error.

The forecast results for precipitation is slightly different than those of temperature and PDSI. The forecast accuracy is measured using the RMSE for each state, which indicates a variability of 11% to 19%. The higher variability can be explained by the fact that precipitation has much stronger long-memory cycles (150 to 300 years) and a fifty-year forecast is only part of the cycle. It is interesting to note that Nevada and Colorado will experience an increase in precipitation in the near future (next 5 to 10 years), however; the long term forecast horizon (10 to 20 years) calls for less precipitation. This indicates that the short forecast horizon for these 2 states is at the end of one cycle and the beginning of the next cycle.

4. Conclusions

The proposed hybrid modeling approach is useful for forecasting drought, temperature, and precipitation at the level of state climate divisions for a span of 1 to 15 years. This process requires the reconstruction of past climate variables that are stationary and stochastically self-similar with quantile regression modeling used to facilitate quantization of forecast uncertainty. The application to southwestern states provided reconstructions of past (0-2009) climate records exhibiting temporal paleoclimatic features, such as the Medieval Warm Period and Little Ice Age. Independent performance testing using modern (2012) observations averaged over appropriate climate divisions demonstrated good correspondence to median forecasts.

This finding supports the 50 years of forecasting (2010–2060) to assist managers in formulating decisions, potential mitigating strategies, and policy associated with future, uncertain climate-change. Differences among median forecasts and observations from other climate divisions demonstrate the heterogeneous nature of climate variables within each state. This finding supports future improvements in forecasting by introducing additional paleoclimatic data associated with grid nodes crossing all of the state climate divisions. Because the proposed hybrid approach can be extended to any spatiotemporal scale providing self-similarity exists, forecasting has the possibility to address economic, political, and social aspects affecting various sectors of society. As with any data-driven approach, the introduction of additional related information can be expected to further reduce the uncertainty.

While the approach presented was geared toward climate change, the work here can be extended to any type of time series analysis. It represents a superior method to

traditional accuracy measures of forecasting. That is, traditional measures are tuned to account for errors in the model, however; the method presented here accounts for the uncertainty of the entire modelling process. The method can be applied to time series and spatial analysis in fields such as economics, ecology, and biology. Beyond the novel method presented for forecasting uncertainty, the fractal information dimension transformation of data can be used to find changes to time series. Current research in the area concentrates on the correlation of volcanic activities to climate shocks found in the fractal transformation of 2000 years of data presented in this research (Friedel and Akbari Esfahani, 2013).

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